# Discussion paper for CIE review of the eastern Bering Sea walleye pollock stock assessment

**May 2025, CIE review**

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# Executive Summary

The work presented below was based on the same data sets used in the 2024 assessment. The first task was to update the model runs to include the 2023 data. The model runs were updated and the results are presented below. The model runs were then used to compare the FT-NIR data with the traditional data sources (BTS and ATS). The results of these comparisons are presented below. The model fits to the data are presented in the tables at the end of the document. The first section steers towards the R library used for processing model results for EBS pollock. The second section presents the comparisons of the FT-NIR data with the traditional data sources. The third section compares results of different software platforms (e.g., SS3, CEA, and AMAK) for the assessment. The fourth section shows results comparing newer model-based estimates from survey data.

# R Library for assessment compilations (ebswp)

The `ebswp` library is a collection of functions for compiling and analyzing results from the stock assessment models for the eastern Bering Sea walleye pollock (Ianelli (2025)). The library is designed to work with the ADMB code and depends on a consistent format for the report files generated. The link in the reference provides access to some notes on applications as vignettes to the package (e.g., . here).

# Comparing FT-NIR compositions with traditional microscope age estimations

The fish age and growth lab at the AFSC have endeavored to expand on new technologies (Helser et al. (2019)).

## FT-NIRS data consistency comparisons

Examining the abundance-at-age (for design-based estimates of the bottom-trawl age compositions) we show that in general, the TMA estimates show a high level of consistency by cohort (Figure 1). For these same data, when applied to the FTNIRs age-estimation method, the consistency was lower (Figure 2). The extent that this is a concern is unknown.

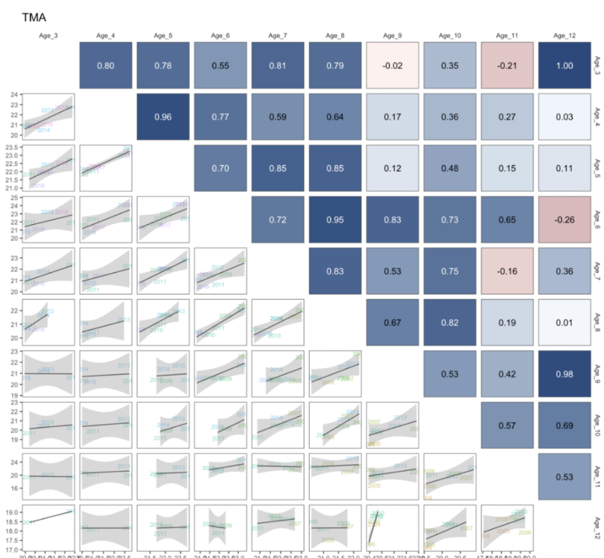


Figure 1: Cohort consistency over different ages based on log-abundance from the bottom-trawl survey data for pollock in recent years for traditional microscope age estimates. The cohorts are shown in the labels.

When preparing the data for the assessment, we plotted out the proportions-at-age for the TMA data and noted the differences in the estimates using the FTNIRs age estimates (Figure 3,

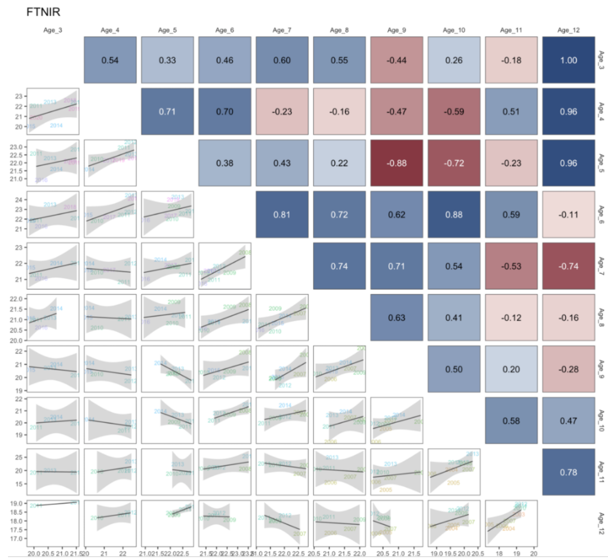

Figure 2: Cohort consistency over different ages based on log-abundance from the bottom-trawl survey data for pollock in recent years for FTNIRs age estimates. The cohorts are shown in the labels.

Figure 4, and Figure 5). Other preparations involved estimating the age-determination errors for each method following that of Punt et al. (2008). For the FTNIRs data, we optionally estimated separate age-estimation error matrices by gear type. The appearance of the fleet-aggregated and fleet-specific input data files is shown in Figure 6.

For the assessment model, the alternatives are shown based on the coverage of data and availability among types of ageing is shown in Figure 7. These figures are intended to show what data are available for the model testing and the different ways of treating age determination error matrices.

## Comparisons of fit to FT-NIRS models

The layout of model runs comparing the traditional microscope ages (TMA) with the FT-NIRS ages is shown in Table 1. The models are run with the FT-NIRS data with different age-error matrix specficiations. The model fits to the data are shown in Table 2. The table shows the goodness-of-fit

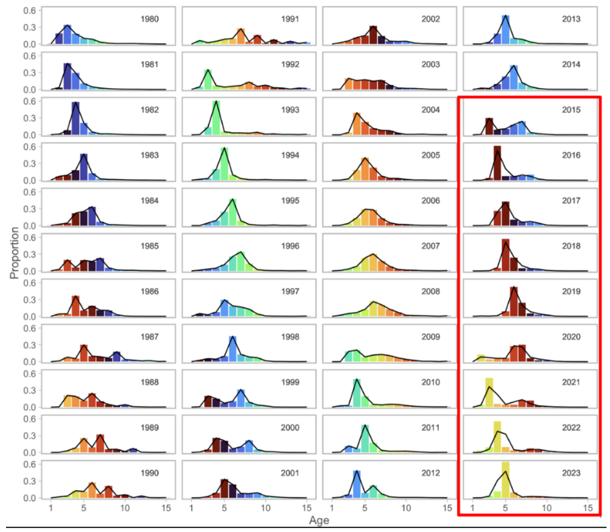

Figure 3: Proportions at age for recent years for the fishery data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

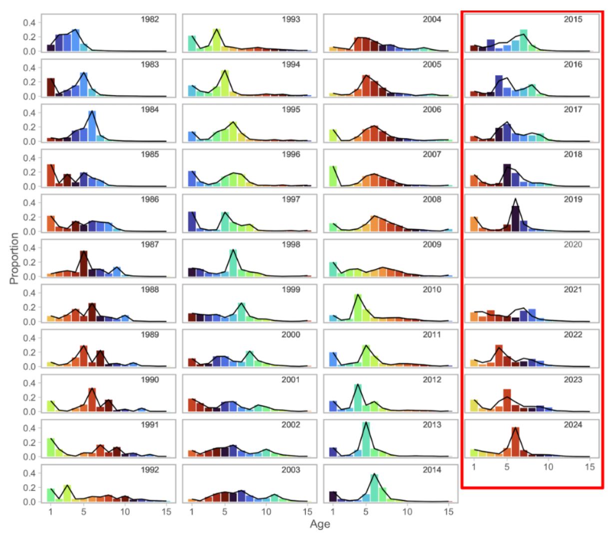

Figure 4: Proportions at age for recent years for the bottom-trawl survey data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

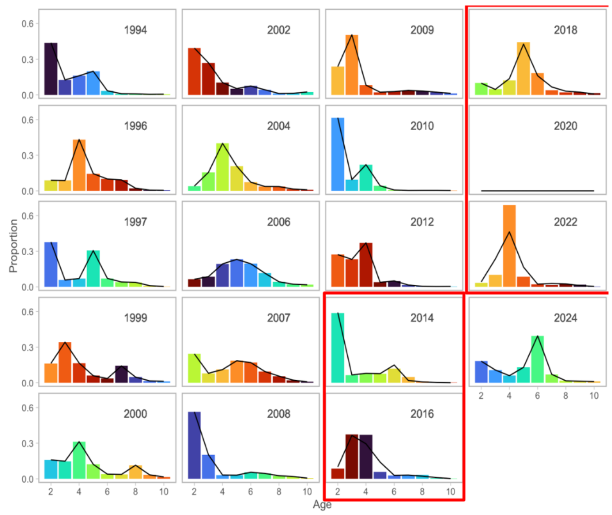


Figure 5: Proportions at age for recent years for the acoustic-trawl survey data where the columns represent the traditional microscope age estimates (TMA) compared to the FTNIRs age estimates (lines for the red-outlined panels).

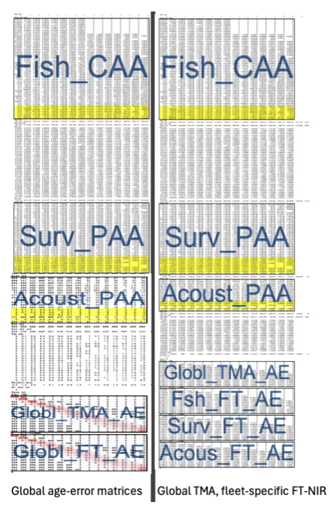


Figure 6: Schematic of input datafile showing the breakout of the fleet-specific FTNIRs age-error matrices (right panel) compared to that where the global FTNIRs age-error matrices are used (left panel).

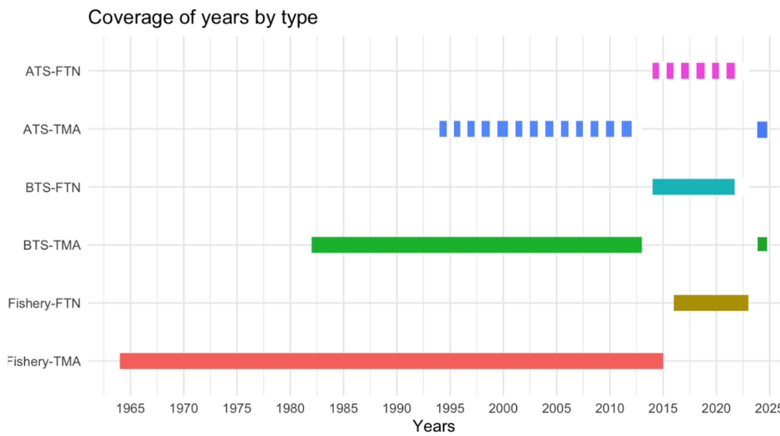


Figure 7: Data extent by age composition for the models with FTNIRs age estimates. The data types are for the Fishery, bottom trawl survey (BTS), and acoustic trawl survey (ATS) and the years of coverage are shown in the x-axis.

Table 1: Configuration of different model options testing the impact of different treatment of the age composition data and ageing error.

| | Model description | |
|---|---|---|
| Name | Age type | Age-error |
| Base | TMA | Global TMA |
| FT-NIR 1 | TMA + FT-NIR | Global TMA, global FT-NIR |
| FT-NIR 2 | TMA + FT-NIR | Global TMA, disaggregated FT-NIR |

Table 2: Goodness-of-fit measures to primary data for different data and age-error applications. RMSE=root-mean square log errors, NLL=negative log-likelihood, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data). See text for incremental model descriptions.

| Component | Base | FT-NIR fleets aggregated | FT-NIR fleet-specific |
|---|---|---|---|
| RMSE BTS | 0.150 | 0.151 | 0.152 |
| RMSE ATS | 0.167 | 0.163 | 0.158 |
| RMSE AVO | 0.216 | 0.217 | 0.215 |
| RMSE CPUE | 0.097 | 0.097 | 0.097 |
| SDNR BTS | 0.890 | 0.900 | 0.880 |
| SDNR ATS | 0.930 | 0.910 | 0.880 |
| SDNR AVO | 0.950 | 0.960 | 0.940 |
| Eff. N Fishery | 358 | 336 | 381 |
| Eff. N BTS | 159 | 148 | 153 |
| Eff. N ATS | 233 | 221 | 233 |
| Catch NLL | 3 | 4 | 3 |
| BTS NLL | 24 | 24 | 23 |
| ATS NLL | 8 | 7 | 7 |
| ATS Age1 NLL | 7 | 8 | 9 |
| AVO NLL | 8 | 8 | 8 |
| Fish Age NLL | 417 | 559 | 614 |
| BTS Age NLL | 286 | 330 | 313 |
| ATS Age NLL | 43 | 42 | 41 |
| NLL selectivity | 213 | 216 | 215 |
| NLL Priors | 20 | 20 | 20 |
| Data NLL | 807 | 995 | 1,032 |
| Total NLL | 1110.900 | 1298.930 | 1332.810 |

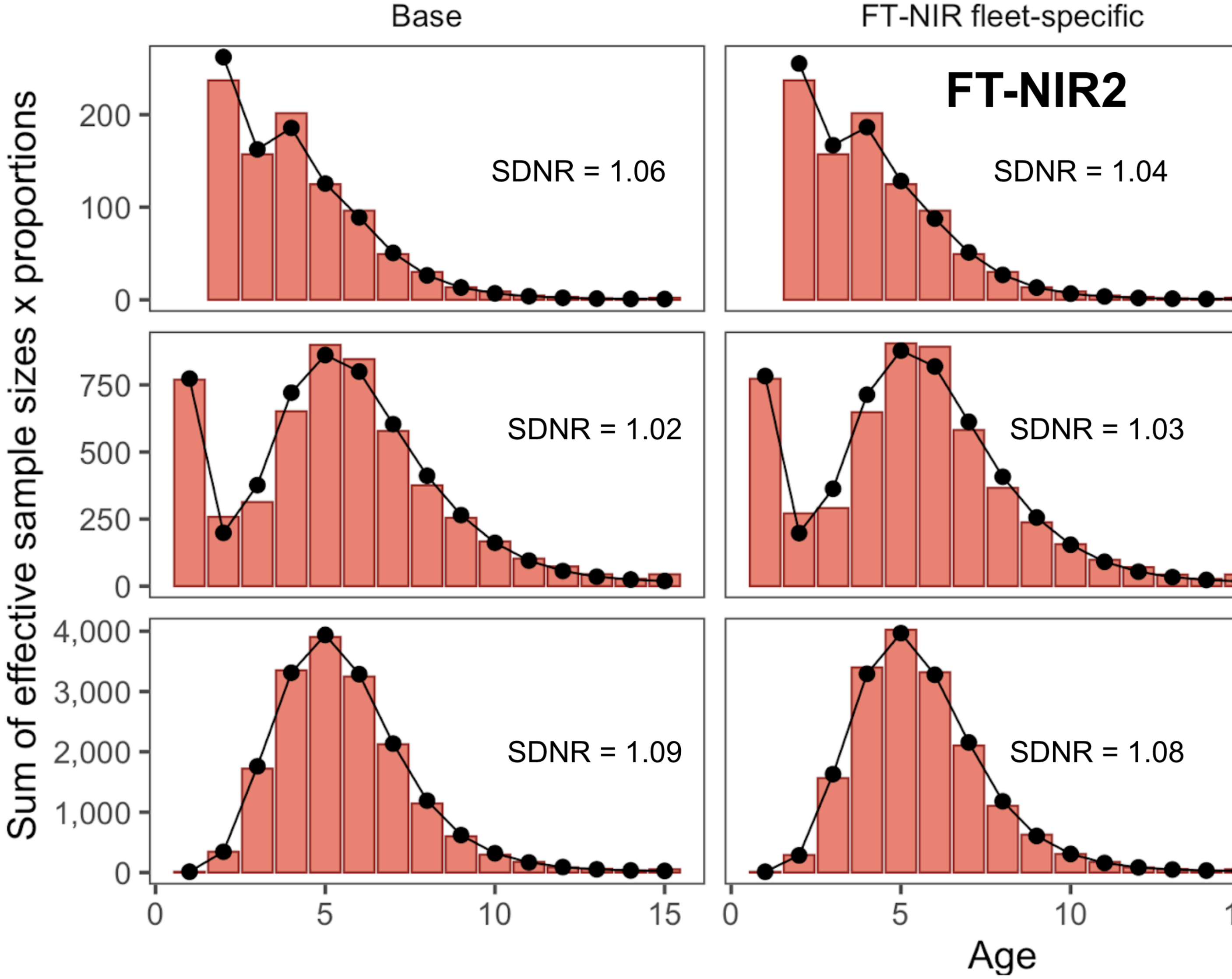


Figure 8: One-step ahead residuals for different model specifications for the fishery composition data

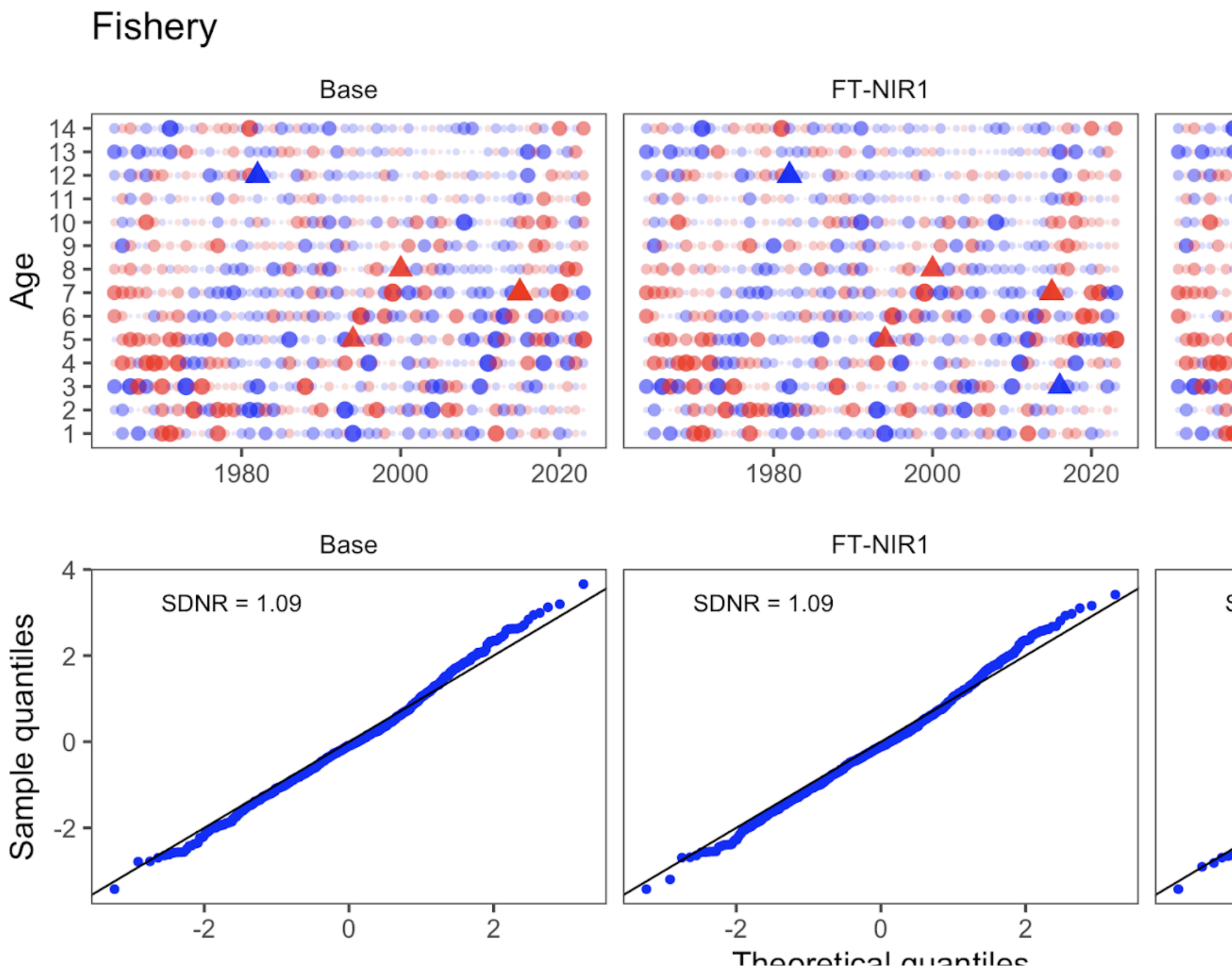


Figure 9: One-step ahead residuals for different model specifications for the fishery composition data

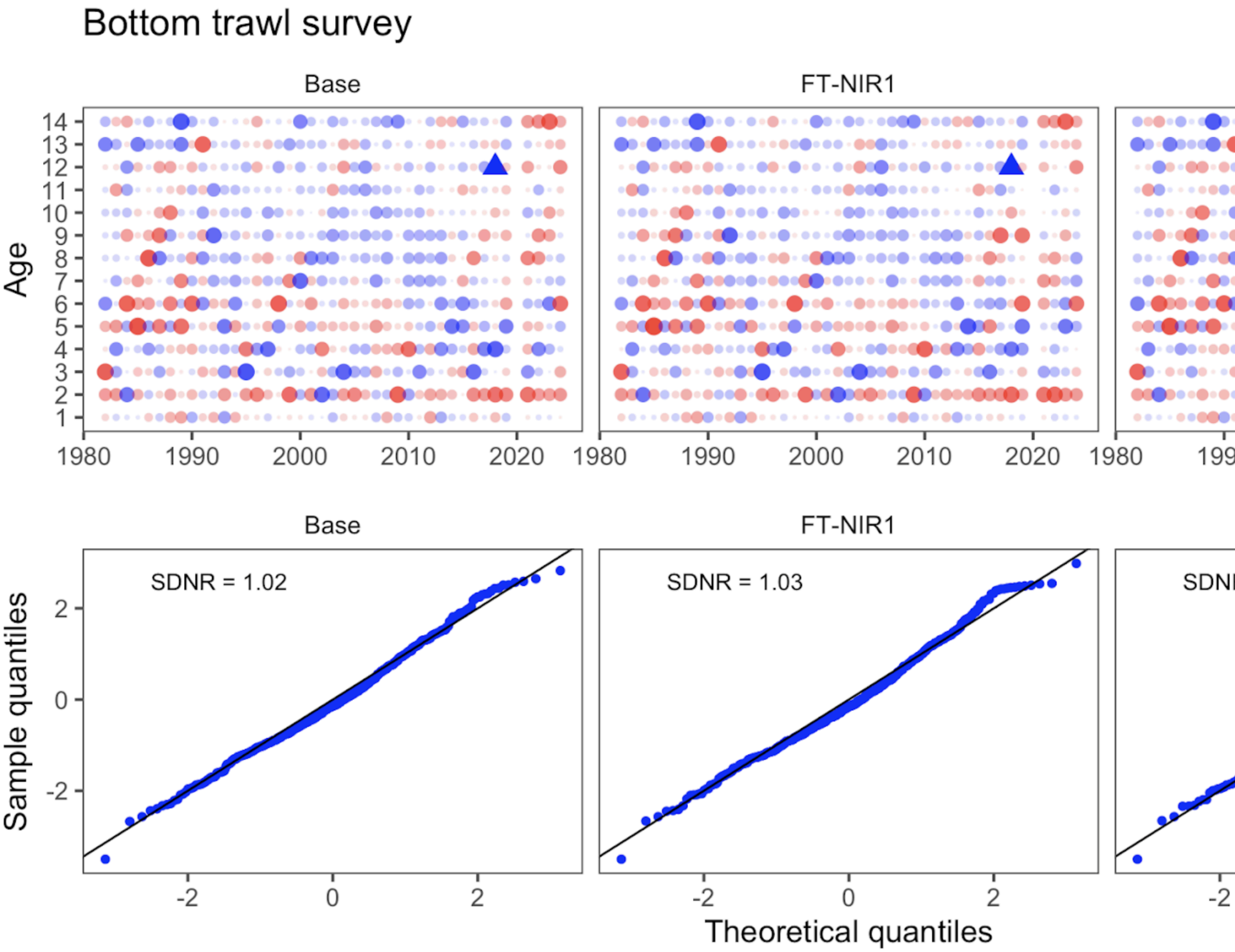


Figure 10: One-step ahead residuals for different model specifications for the bottom-trawl survey composition data

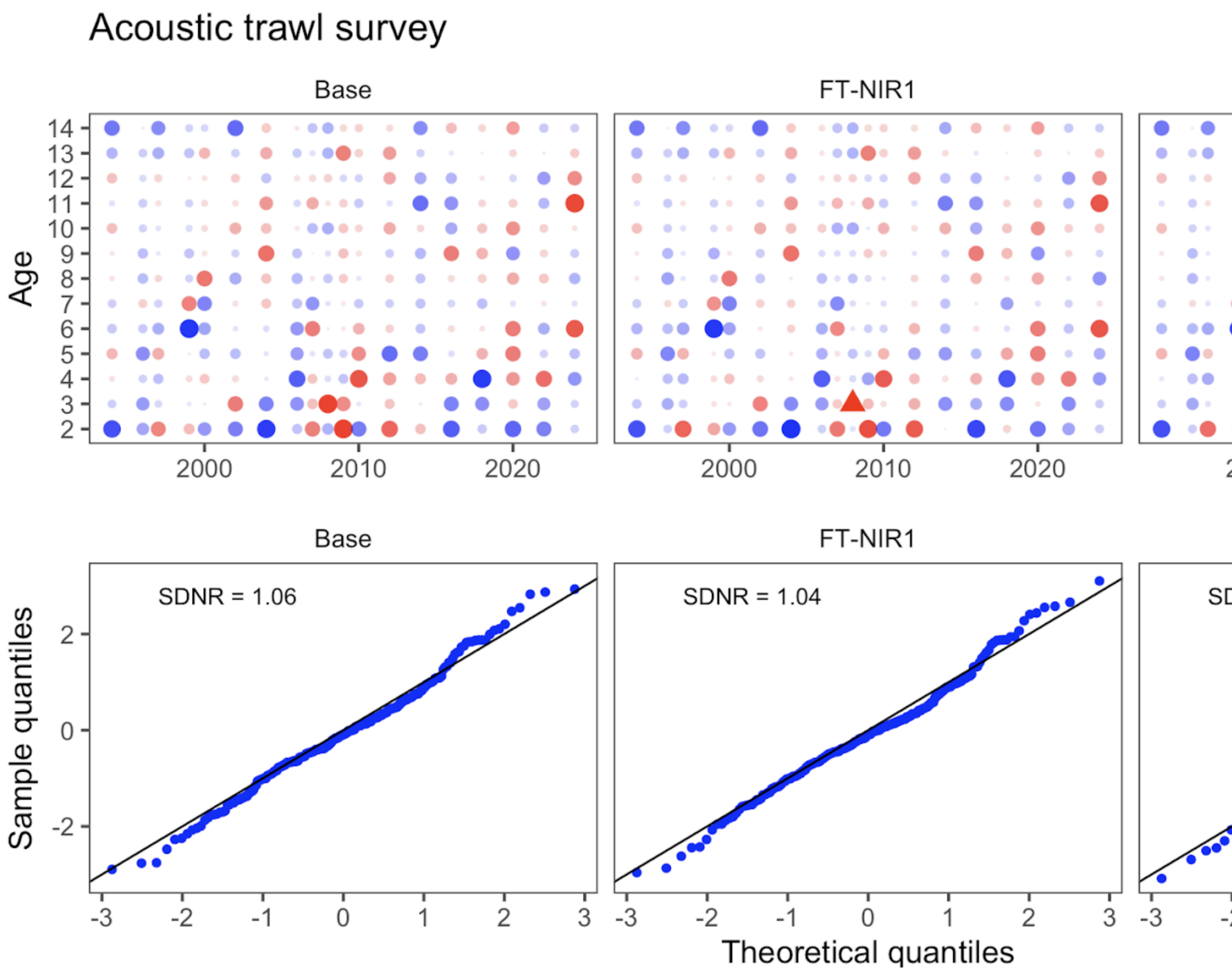


Figure 11: One-step ahead residuals for different model specifications for the acoustic-trawl survey composition data

Table 3: Goodness-of-fit measures to primary data for different data and age-error applications. RMSE=root-mean square log errors, NLL=negative log-likelihood, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data). See text for incremental model descriptions.

| Component | Base | FT-NIR fleets aggregated | FT-NIR fleet-specific |
|---|---|---|---|
| $B_{2025}$ | 2,800 | 2,500 | 2,700 |
| $CV_{B_{2025}}$ | 0.14 | 0.14 | 0.14 |
| $B_{MSY}$ | 1,667 | 1,652 | 1,670 |
| $CV_{B_{MSY}}$ | 0.23 | 0.24 | 0.23 |
| $B_{2025}/B_{MSY}$ | 165% | 153% | 162% |
| $B_0$ | 4,318 | 4,360 | 4,436 |
| $B_{35\%}$ | 1,935 | 1,898 | 1,931 |
| SPR rate at $F_{MSY}$ | 31% | 30% | 29% |
| Steepness | 0.63 | 0.64 | 0.64 |
| Est. $B_{2024}/B_{2024,nofishing}$ | 0.65 | 0.62 | 0.64 |
| $B_{2024}/B_{MSY}$ | 182% | 169% | 178% |

# Leave-one-out survey sensitivity analysis

Here we present model runs that downweight one or more sources of survey data at a time. Table 1 shows the combinations of data types included for this analysis, which includes six model runs: (1) downweight all Acoustic Trawl Survey (ATS) data (age 1 survey index, age 2+ survey biomass index, and age composition data), (2) downweight the ATS age composition data, (3) downweight the Acoustic Vessels of Opportunity (AVO) data (survey biomass index), (4) downweight all Eastern Bering Sea Trawl Survey (BTS) data (age 1+ survey biomass index and age composition data), (5) downweight the BTS age composition data, and (6) downweight all survey biomass indices (BTS, ATS, and AVO survey biomass indices). The ATS age 1 data are included as a separate survey index in the assessment model, while BTS age 1 data are included with the age composition data in the multinomial objective function component.

Among these models, downweighting the BTS data (both solely downweighting the BTS age composition data and downweighting all BTS data) led to changes in trends over time in estimated spawning biomass Figure 12 . The model estimates of recruitment were variable across sensitivity runs, but estimates differed the most from the 2024 accepted model when all BTS data were downweighted Figure 13. Estimates of BTS survey biomass fit the BTS data very poorly in most years when the BTS data were downweighted (Figure 14). Additionally, the fits to the BTS survey biomass were affected by downweighting the BTS age composition data (without downweighting the BTS survey index). The model was relatively insensitive to other data being excluded, and fits to the ATS and AVO indices were relatively insensitive to downweighting the BTS data (see Figure 14, Figure 15, and Figure 16). The root mean square error (RMSE) in Table 2 for the model run that downweighted all BTS data was large and the and effective sample size (Eff. N) was low as compared to the 2024 accepted model.

Table 1: Configuration of different model options testing the impact of different data components

| | Model description | | | | | |
|---|---|---|---|---|---|---|
| Dataset | Drop all BTS | Drop all ATS | Drop all indices | Drop AVO | Drop BTS Age composition | Drop ATS Ag |
| BTS age 2+ Index | X | | X | | | |
| ATS age 2+ Index | | X | X | | | |
| ATS age 1 Index | | X | X | | | |
| AVO Index | | | X | | | |
| BTS comps | X | | | | X | |
| ATS comps | | X | | | | X |

Additionally, downweighting solely the BTS age composition data led to low effective sample size for the BTS.

The importance of the survey data can be seen in some of the reference calculations where the uncertainty increases dramatically and the stock-status perception was quite different across sensitivity runs Table 3. Most dramatically, dropping all survey indices led to a CV in "$B_{MSY}$" estimates of 1.26 as compared to 0.13 in the 2024 accepted model.

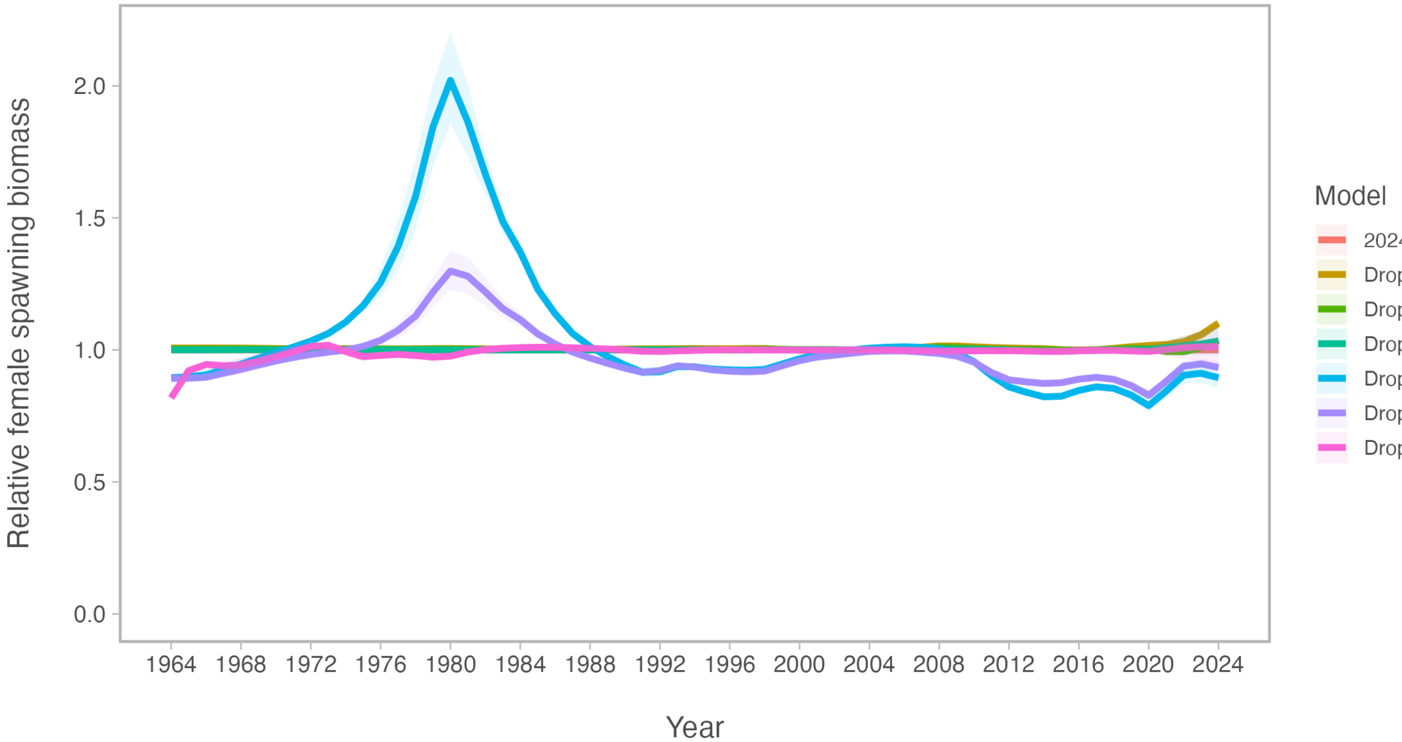


Figure 12: Comparison of spawning biomass (SSB) relative to the 2024 accepted model for models that downweight one data source at a time

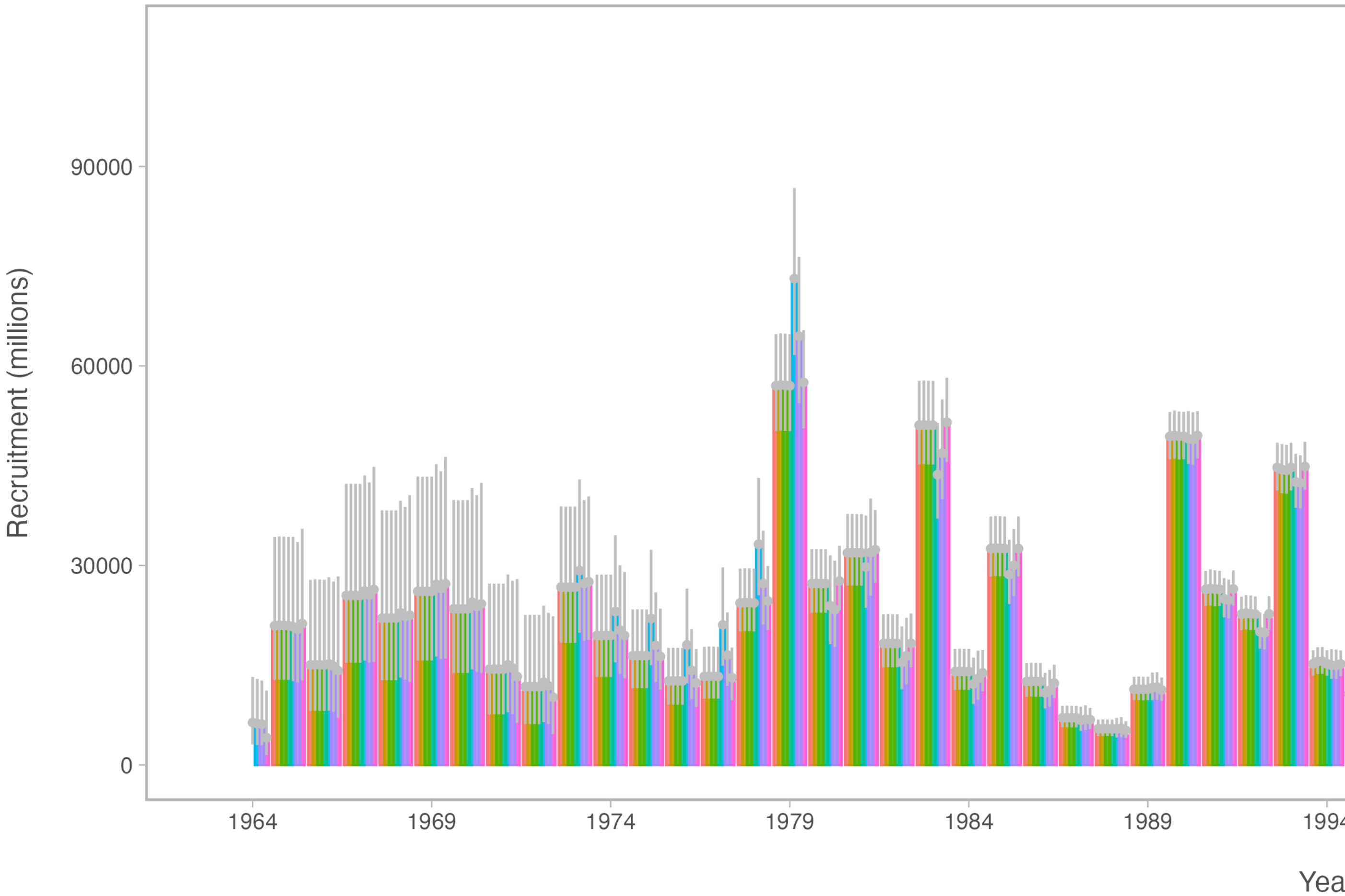


Figure 13: Comparison of recruitment estimates for models that downweight one data source at a time

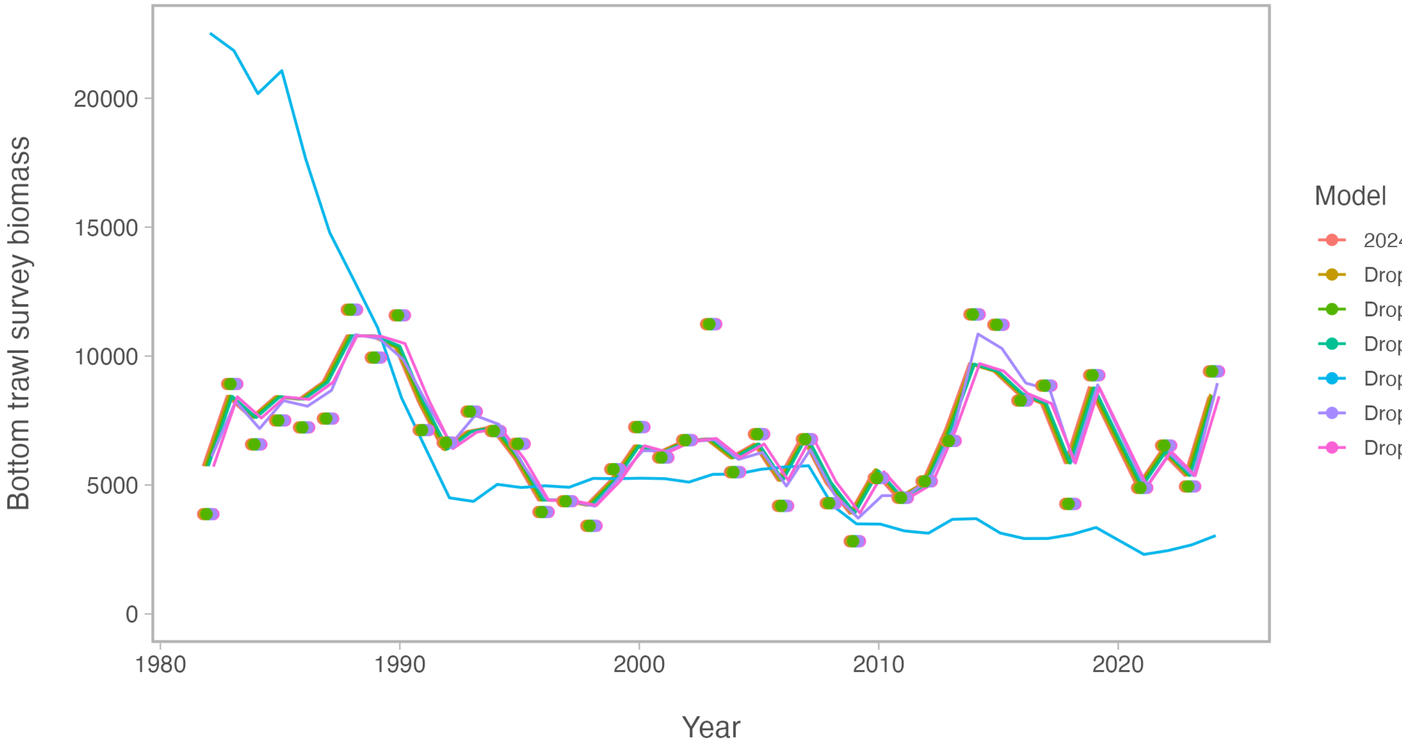


Figure 14: Comparison of fits to the Eastern Bering Sea Bottom Trawl Survey (BTS) biomass index for models that downweight one data source at a time

Table 2: Goodness-of-fit measures to primary data for leave-one-out model sensitivity runs. RMSE=root-mean square log errors, Eff. N=effective sample size for composition data).

| Component | 2024 accepted | Drop BTS | Drop ATS | Drop AVO | Drop indices | Drop BTS age |
|---|---|---|---|---|---|---|
| RMSE BTS | 0.157 | 0.689 | 0.157 | 0.157 | 0.157 | 0.152 |
| RMSE ATS | 0.180 | 0.182 | 0.188 | 0.191 | 0.187 | 0.178 |
| RMSE AVO | 0.229 | 0.237 | 0.238 | 0.241 | 0.238 | 0.231 |
| RMSE CPUE | 0.093 | 0.090 | 0.093 | 0.093 | 0.098 | 0.092 |
| Eff. N Fishery | 1,182 | 1,379 | 1,190 | 1,185 | 1,176 | 1,344 |
| Eff. N BTS | 248 | 95 | 254 | 249 | 245 | 84 |
| Eff. N ATS | 269 | 230 | 164 | 269 | 273 | 228 |

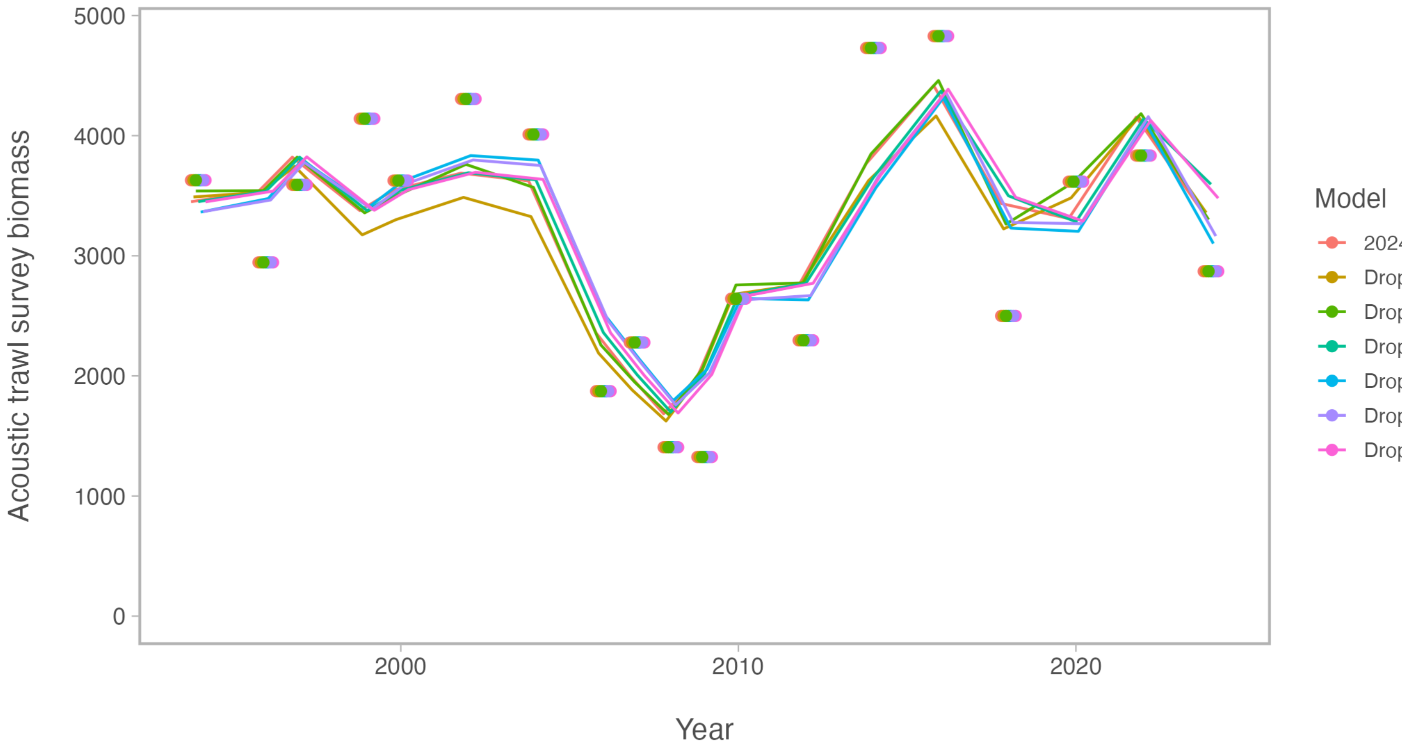


Figure 15: Comparison of fits to the Acoustic Trawl Survey (ATS) biomass index for models that downweight one data source at a time

Table 3: Estimated derived quantities and selected corresponding CVs for leave-one-out sensitivity analysis runs. See text for incremental model descriptions.

| Estimated Quantity | 2024 accepted | Drop BTS | Drop ATS | Drop AVO | Drop indices | Drop |
|---|---|---|---|---|---|---|
| $B_{2025}$ | 3,100 | 2,700 | 3,600 | 3,300 | 3,200 | |
| $CV_{B_{2025}}$ | 0.13 | 0.16 | 0.14 | 0.16 | 0.18 | |
| $B_{MSY}$ | 2,310 | 2,415 | 2,360 | 2,322 | 3,677 | |
| $CV_{B_{MSY}}$ | 0.3 | 0.31 | 0.31 | 0.31 | 1.26 | |
| $B_{2025}/B_{MSY}$ | 135% | 111% | 151% | 141% | 86% | |
| $B_0$ | 5,975 | 6,136 | 6,120 | 6,009 | 8,934 | |
| $B_{35\%}$ | 2,066 | 1,988 | 2,101 | 2,080 | 2,074 | |
| SPR rate at $F_{MSY}$ | 31% | 35% | 31% | 31% | 42% | |
| Steepness | 0.62 | 0.59 | 0.62 | 0.62 | 0.52 | |
| Est. $B_{2024}/B_{2024,nofishing}$ | 0.61 | 0.51 | 0.63 | 0.61 | 0.42 | |
| $B_{2024}/B_{MSY}$ | 147% | 126% | 158% | 151% | 93% | |


Figure 16: Comparison of fits to the Acoustic Vessels of Opportunity (AVO) biomass index for models that downweight one data source at a time

# Incorporating multi-species M-at-age and year estimates, also total uncertainty in acoustics...

A commonly adopted approach is to use estimated natural mortality-at-age and year from multispecies trophic interaction models (e.g., Trijoulet, Fay, and Miller (2020)). The current assessment has an option to include the 2024 CEATTLE estimates of natural mortality (Table 1; Figure 17). This resulted in slightly higher recruitment (Figure 18) lower spawning biomass in the near term (Figure 19). This is due to the higher natural mortality for most ages and years compared to the base 2024 model.

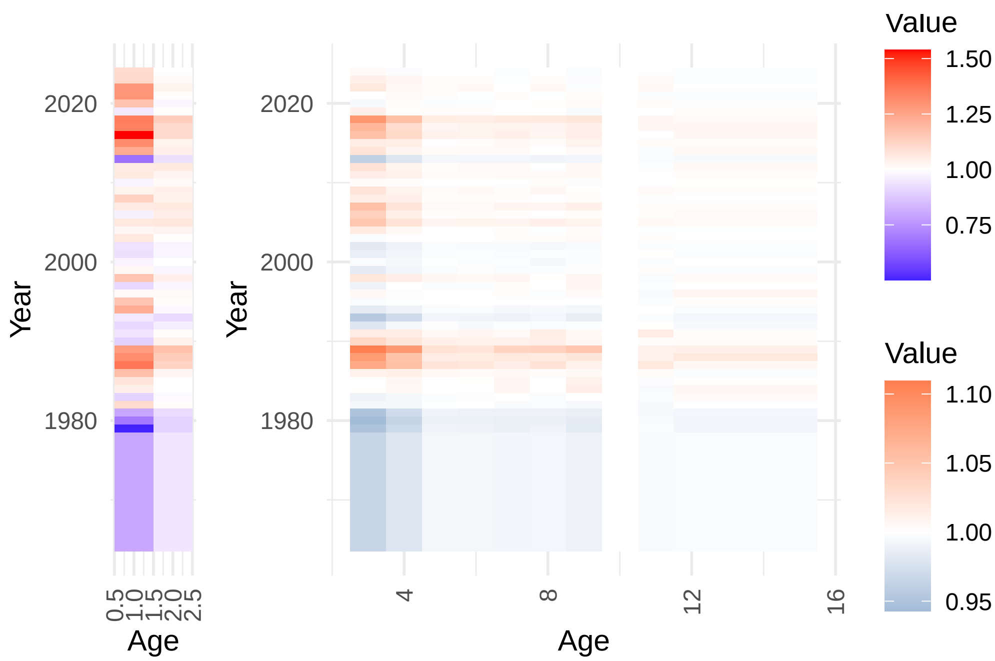


Figure 17: Depiction of the natural mortality (variability from mean value at age) as estimated from CEATTLE.

We also include a comparison of revised acoustic-trawl estimates (Urmy, Ressler, and De Robertis (2025); Figure 20). The natural mortaliity alternative run resulted in generally better fits to most data components whereas the "untuned" run with the Urmy, Ressler, and De Robertis (2025) estimates were generally poorer (Table 2). Results from the Urmy, Ressler, and De Robertis (2025) had minor impact on management-related quantities whereas those

Table 1: Natural mortality-at-age used over ages and time based on the CEATTLE 2024 model.

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1964 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1965 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1966 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1967 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1968 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1969 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1970 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1971 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1972 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1973 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1974 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1975 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1976 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1977 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1978 | 1.024 | 0.341 | 0.320 | 0.311 | 0.303 | 0.304 | 0.303 | 0.304 | 0.304 | 0.3001 |
| 1979 | 0.644 | 0.326 | 0.315 | 0.308 | 0.303 | 0.303 | 0.302 | 0.303 | 0.302 | 0.3001 |
| 1980 | 0.883 | 0.325 | 0.313 | 0.307 | 0.302 | 0.302 | 0.302 | 0.303 | 0.302 | 0.3001 |
| 1981 | 1.022 | 0.333 | 0.315 | 0.309 | 0.303 | 0.303 | 0.303 | 0.303 | 0.303 | 0.3001 |
| 1982 | 1.415 | 0.365 | 0.329 | 0.316 | 0.305 | 0.306 | 0.305 | 0.305 | 0.305 | 0.3001 |
| 1983 | 1.156 | 0.358 | 0.328 | 0.316 | 0.305 | 0.306 | 0.306 | 0.305 | 0.307 | 0.3001 |
| 1984 | 1.342 | 0.363 | 0.332 | 0.320 | 0.305 | 0.306 | 0.308 | 0.306 | 0.312 | 0.3001 |
| 1985 | 1.380 | 0.362 | 0.331 | 0.320 | 0.306 | 0.306 | 0.308 | 0.307 | 0.311 | 0.3001 |
| 1986 | 1.517 | 0.371 | 0.336 | 0.322 | 0.307 | 0.307 | 0.308 | 0.307 | 0.309 | 0.3001 |
| 1987 | 1.756 | 0.406 | 0.356 | 0.335 | 0.312 | 0.312 | 0.310 | 0.314 | 0.311 | 0.3001 |
| 1988 | 1.684 | 0.415 | 0.360 | 0.335 | 0.310 | 0.311 | 0.312 | 0.312 | 0.314 | 0.3001 |
| 1989 | 1.632 | 0.425 | 0.368 | 0.346 | 0.313 | 0.313 | 0.317 | 0.318 | 0.322 | 0.3001 |
| 1990 | 1.145 | 0.377 | 0.343 | 0.325 | 0.309 | 0.310 | 0.309 | 0.311 | 0.309 | 0.3001 |
| 1991 | 1.206 | 0.365 | 0.337 | 0.323 | 0.307 | 0.308 | 0.307 | 0.311 | 0.309 | 0.3001 |
| 1992 | 1.169 | 0.348 | 0.325 | 0.316 | 0.305 | 0.305 | 0.305 | 0.305 | 0.307 | 0.3001 |
| 1993 | 1.220 | 0.332 | 0.316 | 0.309 | 0.303 | 0.303 | 0.303 | 0.304 | 0.303 | 0.3001 |
| 1994 | 1.576 | 0.357 | 0.326 | 0.315 | 0.304 | 0.305 | 0.304 | 0.305 | 0.305 | 0.3001 |
| 1995 | 1.493 | 0.365 | 0.331 | 0.318 | 0.306 | 0.306 | 0.306 | 0.307 | 0.307 | 0.3001 |
| 1996 | 1.306 | 0.367 | 0.334 | 0.318 | 0.305 | 0.306 | 0.307 | 0.306 | 0.309 | 0.3001 |
| 1997 | 1.168 | 0.355 | 0.328 | 0.318 | 0.305 | 0.305 | 0.307 | 0.306 | 0.310 | 0.3001 |
| 1998 | 1.492 | 0.377 | 0.338 | 0.323 | 0.308 | 0.308 | 0.308 | 0.307 | 0.310 | 0.3001 |
| 1999 | 1.311 | 0.355 | 0.326 | 0.316 | 0.304 | 0.306 | 0.305 | 0.306 | 0.307 | 0.3001 |
| 2000 | 1.249 | 0.362 | 0.330 | 0.316 | 0.305 | 0.305 | 0.305 | 0.305 | 0.306 | 0.3001 |
| 2001 | 1.194 | 0.355 | 0.327 | 0.316 | 0.305 | 0.305 | 0.305 | 0.306 | 0.306 | 0.3001 |
| 2002 | 1.199 | 0.353 | 0.325 | 0.315 | 0.304 | 0.305 | 0.305 | 0.305 | 0.306 | 0.3001 |
| 2003 | 1.367 | 0.364 | 0.331 | 0.319 | 0.305 | 0.306 | 0.306 | 0.307 | 0.309 | 0.3001 |
| 2004 | 1.312 | 0.374 | 0.337 | 0.320 | 0.306 | 0.306 | 0.306 | 0.306 | 0.308 | 0.3001 |
| 2005 | 1.366 | 0.387 | 0.348 | 0.326 | 0.308 | 0.309 | 0.309 | 0.310 | 0.310 | 0.3001 |
| 2006 | 1.243 | 0.380 | 0.345 | 0.323 | 0.306 | 0.307 | 0.307 | 0.307 | 0.308 | 0.3001 |
| 2007 | 1.355 | 0.385 | 0.349 | 0.325 | 0.307 | 0.307 | 0.309 | 0.309 | 0.311 | 0.3001 |
| 2008 | 1.450 | 0.377 | 0.336 | 0.322 | 0.306 | 0.307 | 0.307 | 0.307 | 0.308 | 0.3001 |
| 2009 | 1.328 | 0.378 | 0.339 | 0.321 | 0.306 | 0.308 | 0.306 | 0.308 | 0.307 | 0.3001 |
| 2010 | 1.250 | 0.368 | 0.333 | 0.319 | 0.306 | 0.306 | 0.306 | 0.307 | 0.307 | 0.3001 |

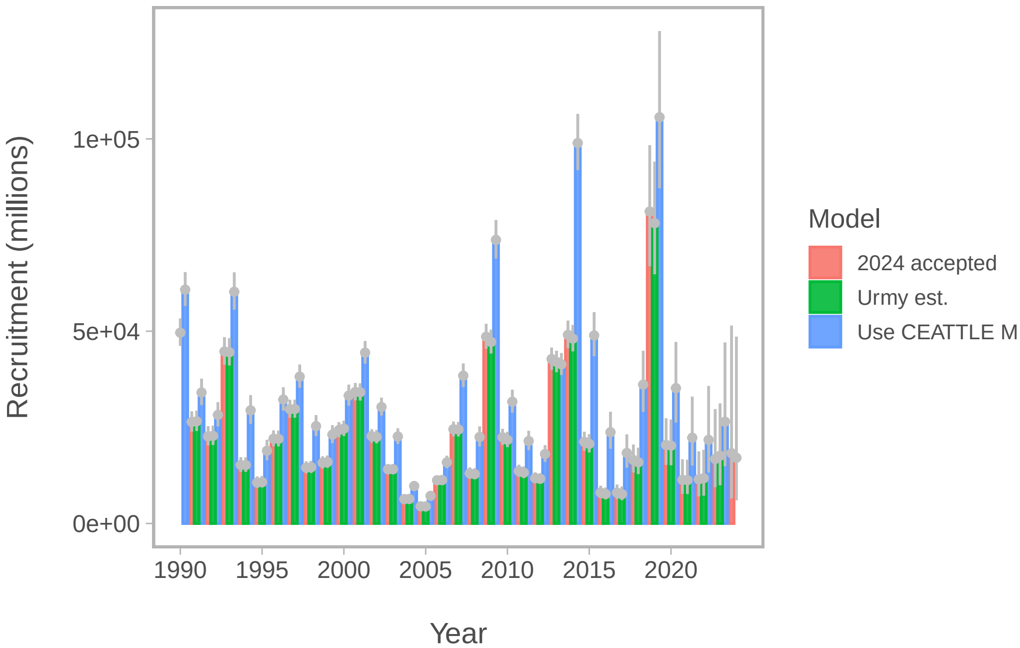


Figure 18: Comparison of recruitment estimates comparing 2024 assessment results with one where M-at-age and year is specified from CEATTLE.

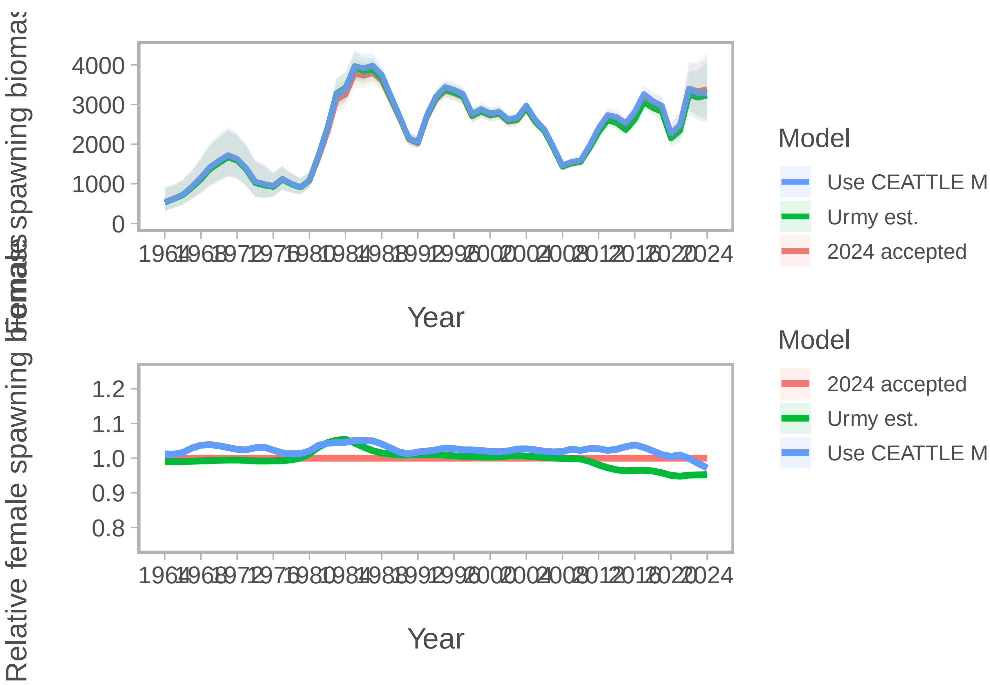


Figure 19: Comparison of spawning biomass (SSB) and SSB relative to the 2024 accepted model for model that uses the M-at-age matrix from CEATTLE.

with the alternative natural mortality specification differed in a number of ways (Table 3). The most notable difference was the estimated $B_{MSY}$ and $B_{35\%}$ values.

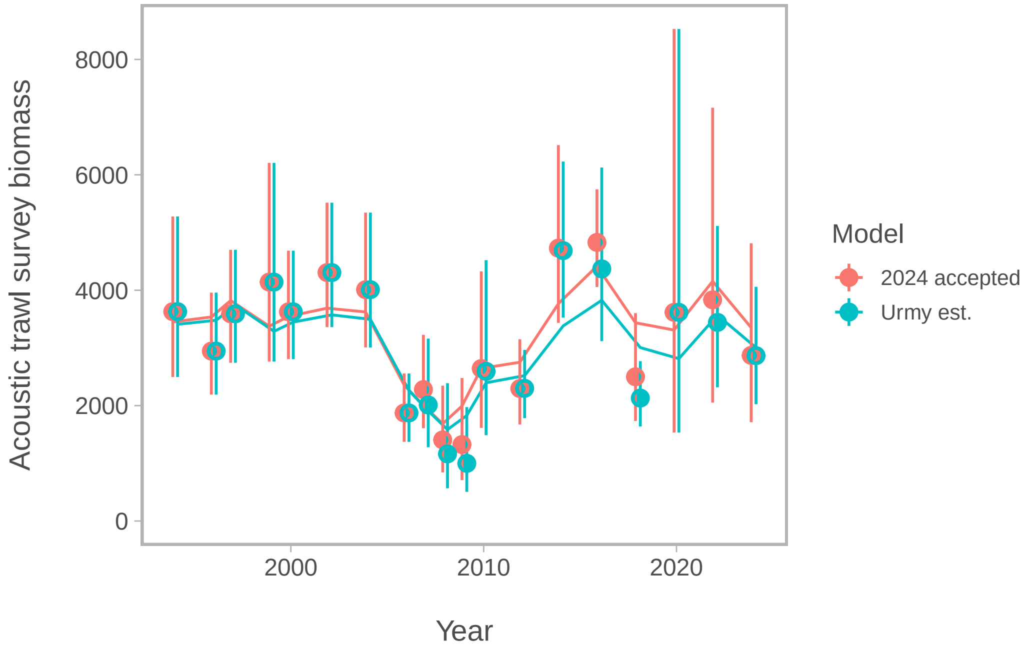


Figure 20: Comparison of fits to the Eastern Bering Sea acoustic-Trawl Survey (ATS) biomass index for the accepted 2024 model and one with revised estimates of index values and uncertainty from Urmy et al., (in prep).

Table 2: Goodness-of-fit measures to primary data for model sensitivity to applying the matrix of M-at-age and year and acoustic uncertainty estimates. RMSE=root-mean square log errors, SDNR=standard deviation of normalized residuals, Eff. N=effective sample size for composition data).

| Component | 2024 accepted | Use CEATTLE M | Urmy est. |
|---|---|---|---|
| RMSE BTS | 0.157 | 0.157 | 0.161 |
| RMSE ATS | 0.180 | 0.179 | 0.226 |
| RMSE AVO | 0.229 | 0.230 | 0.236 |
| RMSE CPUE | 0.093 | 0.093 | 0.093 |
| SDNR BTS | 0.950 | 0.960 | 0.970 |
| SDNR ATS | 1.000 | 0.990 | 1.220 |
| SDNR AVO | 0.980 | 0.960 | 1.020 |
| Eff. N Fishery | 1,182 | 1,176 | 1,188 |
| Eff. N BTS | 248 | 241 | 221 |
| Eff. N ATS | 269 | 267 | 278 |
| Catch NLL | 3 | 3 | 3 |
| BTS NLL | 31 | 31 | 30 |
| ATS NLL | 9 | 9 | 13 |
| ATS Age1 NLL | 11 | 12 | 14 |
| AVO NLL | 9 | 8 | 9 |
| Fish Age NLL | 167 | 168 | 162 |
| BTS Age NLL | 194 | 196 | 205 |
| ATS Age NLL | 30 | 31 | 33 |
| NLL selectivity | 196 | 196 | 199 |
| NLL Priors | 20 | 20 | 20 |
| Data NLL | 465 | 468 | 480 |
| Total NLL | 718 | 713 | 735 |

Table 3: Estimated derived quantities and selected corresponding CVs for sensitivity to natural mortality specification and acoustic uncertainty estimates.

| Estimated Quantity | 2024 accepted | Use CEATTLE M | Urmy est. |
|---|---|---|---|
| $B_{2025}$ | 3,100 | 3,000 | 3,000 |
| $CV_{B_{2025}}$ | 0.13 | 0.13 | 0.13 |
| $B_{MSY}$ | 2,310 | 2,584 | 2,290 |
| $CV_{B_{MSY}}$ | 0.3 | 0.35 | 0.3 |
| $B_{2025}/B_{MSY}$ | 135% | 116% | 131% |
| $B_0$ | 5,975 | 6,568 | 5,929 |
| $B_{35\%}$ | 2,066 | 1,844 | 2,057 |
| SPR rate at $F_{MSY}$ | 31% | 35% | 31% |
| Steepness | 0.62 | 0.59 | 0.62 |
| Est. $B_{2024}/B_{2024,nofishing}$ | 0.61 | 0.54 | 0.59 |
| $B_{2024}/B_{MSY}$ | 147% | 127% | 141% |

# Alternative assessment software

## Stock-synthesis 3

Stock synthesis 3 (SS3) is a well establish framework and is common in many diverse settings. (Methot and Wetzel (2013)) For EBS pollock, we extended an initial framework based on the configurations used for Pacific hake (Grandin et al. (2024)) since there are many similarities in the type of data available. Namely, empirical weight-at-age, acoustic survey data, and indices of 1-year olds.

Within the SS3 framework, we evaluated different configurations for fishery selectivity variability (Figure 21). These showed the impact on the uncertainty and stock trends (Figure 22 for SSB and Figure 23 for recruitment).

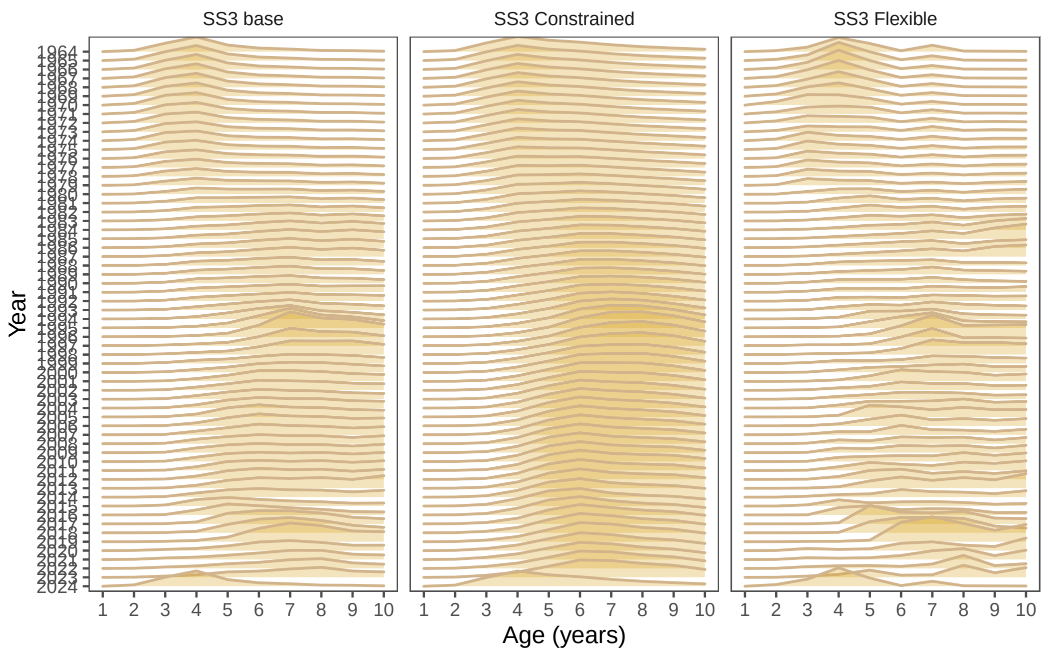


Figure 21: Model results comparing three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

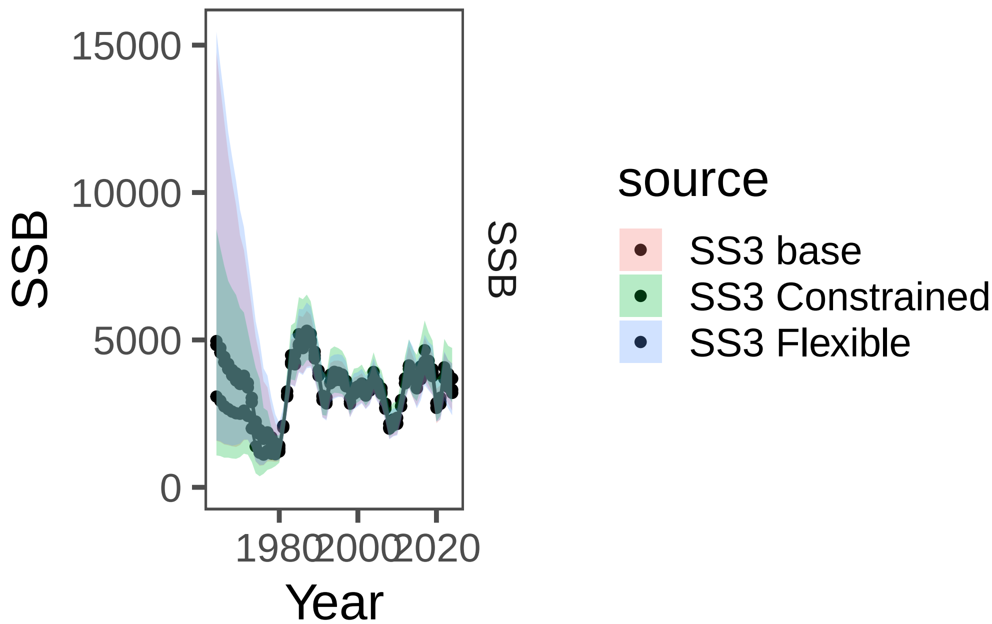


Figure 22: Model results comparing the time-series for three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

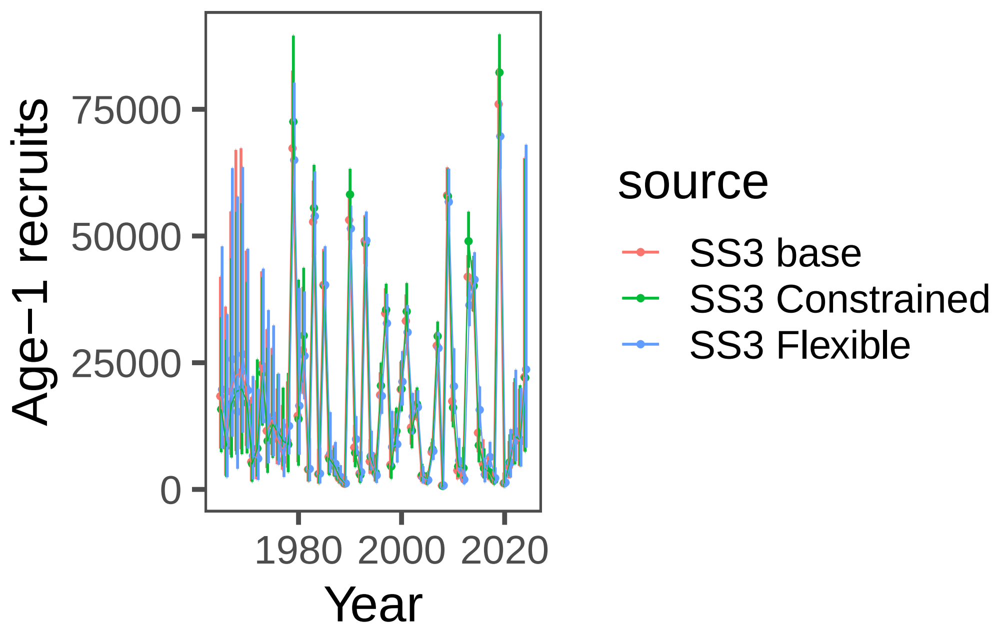


Figure 23: Model results comparing the time-series for three different SS3 model configurations. The models are the base model (Base), the constrained model (Constrained), and the flexible model (Flexible).

Comparing the "base" SS3 model with the pollock model showed different patterns in the early periods for selectivity (Figure 24) but slight differences in SSB and recruitment estimates (Figure 25 and Figure 26).

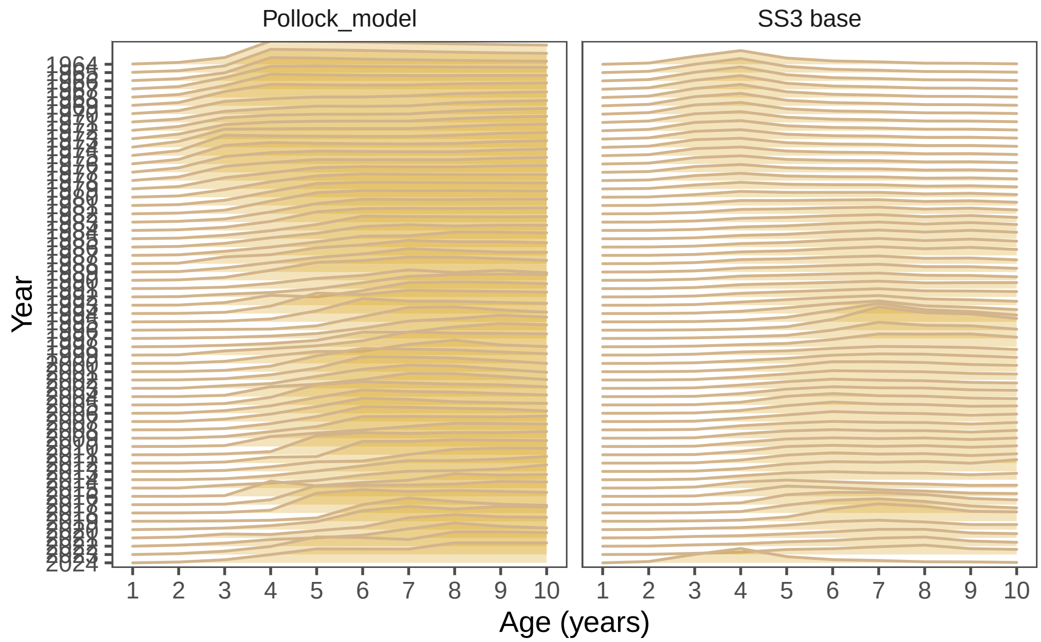


Figure 24: Model results comparing selectivity estimates between the base SS3 model configuration and the pollock model accepted in 2024.

## SAM

The Stock Assessment Model (SAM) is a state-space model written in TMB (Nielsen and Berg (2014), Nielsen et al. (2021)). We attempted to configure the settings to have similar levels of flexibility to the reference model used for catch advice in 2024 and the settings and results can be found here). Comparing "selectivity" in the SAM model to the reference model, we see broad similarities but slightly lower inter-annual variability in the SAM run (Figure 27).

The spawning biomass trend largely overlaps in the uncertainty between the SAM and pollock model (Figure 28). The recruitment estimates are similar as well (Figure 29).

```
       type   source          se      value Year
1    Biomass CEA, FE    598.17756  3439.3743 1964
2    Biomass CEA, FE    527.87773  4308.0457 1965
3    Biomass CEA, FE    489.70388  4912.3713 1966
4    Biomass CEA, FE    494.03978  6013.3280 1967
```

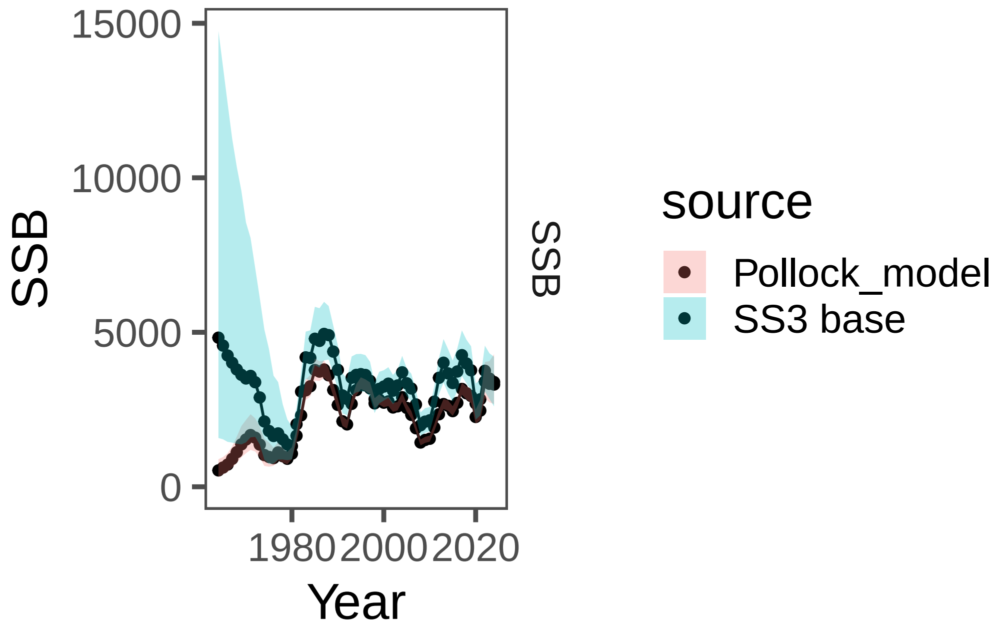


Figure 25: Model results comparing the base SS3 model configuration with the model used for pollock in 2024.

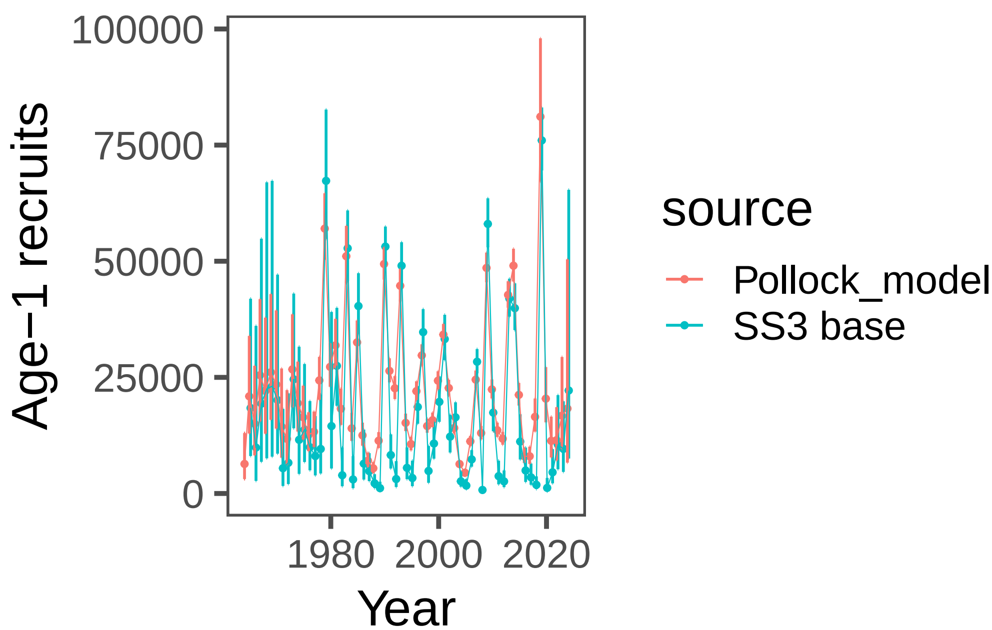


Figure 26: Model results comparing the base SS3 model configuration with the model used for pollock in 2024.

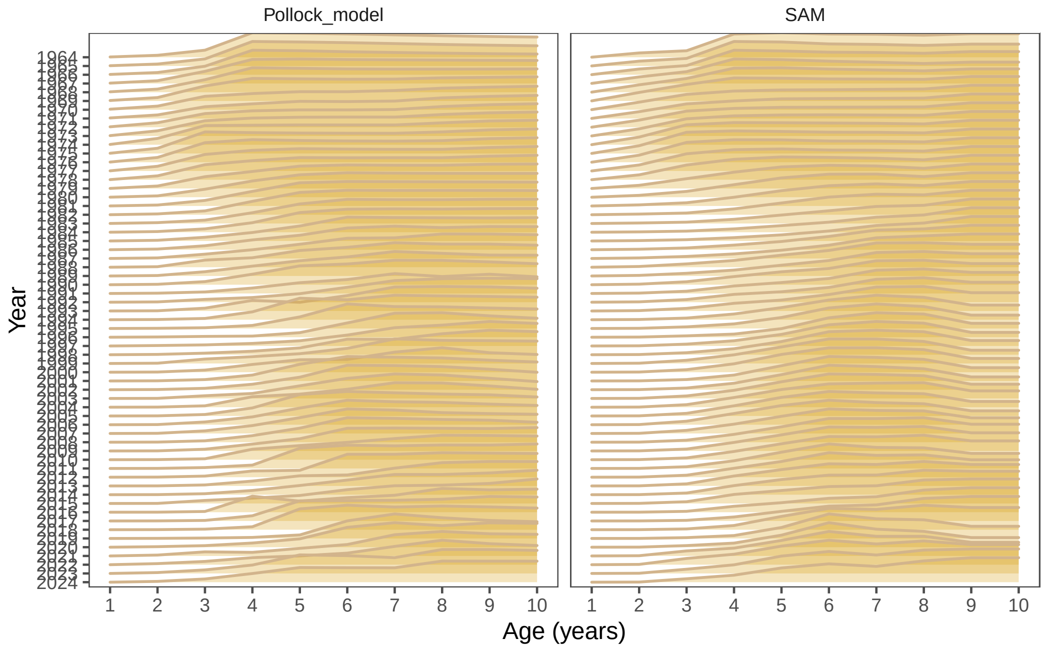


Figure 27: Model results comparing last year's model with the SAM model.

| | | | | |
|---|---|---|---|---|
| 5 | Biomass CEA, FE | 532.92400 | 6969.1707 | 1968 |
| 6 | Biomass CEA, FE | 536.81933 | 7701.4589 | 1969 |
| 7 | Biomass CEA, FE | 557.82544 | 8534.3333 | 1970 |
| 8 | Biomass CEA, FE | 549.78156 | 8506.2570 | 1971 |
| 9 | Biomass CEA, FE | 525.71007 | 7769.6392 | 1972 |
| 10 | Biomass CEA, FE | 469.16874 | 6977.6100 | 1973 |
| 11 | Biomass CEA, FE | 419.40173 | 5905.0511 | 1974 |
| 12 | Biomass CEA, FE | 361.17764 | 5026.7801 | 1975 |
| 13 | Biomass CEA, FE | 296.02148 | 4475.7364 | 1976 |
| 14 | Biomass CEA, FE | 248.40359 | 4103.9810 | 1977 |
| 15 | Biomass CEA, FE | 229.75066 | 4303.2954 | 1978 |
| 16 | Biomass CEA, FE | 212.61136 | 5517.9693 | 1979 |
| 17 | Biomass CEA, FE | 204.64407 | 6322.2711 | 1980 |
| 18 | Biomass CEA, FE | 231.25071 | 7922.8261 | 1981 |
| 19 | Biomass CEA, FE | 284.58651 | 9567.1556 | 1982 |
| 20 | Biomass CEA, FE | 328.12244 | 11015.7151 | 1983 |
| 21 | Biomass CEA, FE | 330.20065 | 10806.5357 | 1984 |
| 22 | Biomass CEA, FE | 386.68764 | 12882.0868 | 1985 |
| 23 | Biomass CEA, FE | 362.30880 | 11770.4904 | 1986 |
| 24 | Biomass CEA, FE | 325.01055 | 11744.3525 | 1987 |
| 25 | Biomass CEA, FE | 298.67476 | 10734.6996 | 1988 |
| 26 | Biomass CEA, FE | 271.66492 | 9254.7113 | 1989 |

| | | | | | |
|---|---|---|---|---|---|
| 27 | Biomass | CEA, FE | 249.10075 | 8901.5154 | 1990 |
| 28 | Biomass | CEA, FE | 262.00036 | 9884.7445 | 1991 |
| 29 | Biomass | CEA, FE | 267.66529 | 10430.5415 | 1992 |
| 30 | Biomass | CEA, FE | 271.21434 | 10603.9665 | 1993 |
| 31 | Biomass | CEA, FE | 267.77579 | 10405.8978 | 1994 |
| 32 | Biomass | CEA, FE | 230.25118 | 8469.3684 | 1995 |
| 33 | Biomass | CEA, FE | 235.26339 | 8549.1984 | 1996 |
| 34 | Biomass | CEA, FE | 218.96539 | 7943.5994 | 1997 |
| 35 | Biomass | CEA, FE | 221.05305 | 8155.2409 | 1998 |
| 36 | Biomass | CEA, FE | 223.92060 | 8514.3632 | 1999 |
| 37 | Biomass | CEA, FE | 232.45297 | 9191.0886 | 2000 |
| 38 | Biomass | CEA, FE | 228.81074 | 9254.6976 | 2001 |
| 39 | Biomass | CEA, FE | 234.93795 | 9913.1315 | 2002 |
| 40 | Biomass | CEA, FE | 261.17480 | 11416.5248 | 2003 |
| 41 | Biomass | CEA, FE | 248.90825 | 10603.4771 | 2004 |
| 42 | Biomass | CEA, FE | 216.65365 | 8781.8837 | 2005 |
| 43 | Biomass | CEA, FE | 198.33784 | 7481.9211 | 2006 |
| 44 | Biomass | CEA, FE | 194.07004 | 7033.9708 | 2007 |
| 45 | Biomass | CEA, FE | 166.34360 | 5747.1408 | 2008 |
| 46 | Biomass | CEA, FE | 195.92949 | 6981.7691 | 2009 |
| 47 | Biomass | CEA, FE | 239.21780 | 9168.7340 | 2010 |
| 48 | Biomass | CEA, FE | 257.68739 | 10059.8552 | 2011 |
| 49 | Biomass | CEA, FE | 272.89309 | 10253.5790 | 2012 |
| 50 | Biomass | CEA, FE | 302.18755 | 11181.0921 | 2013 |
| 51 | Biomass | CEA, FE | 311.24082 | 12050.4492 | 2014 |
| 52 | Biomass | CEA, FE | 324.71242 | 13024.1317 | 2015 |
| 53 | Biomass | CEA, FE | 341.68890 | 13764.8013 | 2016 |
| 54 | Biomass | CEA, FE | 324.71644 | 12420.6702 | 2017 |
| 55 | Biomass | CEA, FE | 296.03558 | 10468.6550 | 2018 |
| 56 | Biomass | CEA, FE | 300.44402 | 10687.9255 | 2019 |
| 57 | Biomass | CEA, FE | 316.67426 | 10874.2267 | 2020 |
| 58 | Biomass | CEA, FE | 334.30851 | 10410.3203 | 2021 |
| 59 | Biomass | CEA, FE | 408.94727 | 11265.9671 | 2022 |
| 60 | Biomass | CEA, FE | 421.92922 | 10146.7683 | 2023 |
| 61 | Biomass | CEA, FE | 543.08471 | 9765.4338 | 2024 |
| 62 | SSB | CEA, FE | 190.87167 | 863.5953 | 1964 |
| 63 | SSB | CEA, FE | 167.78091 | 1070.9124 | 1965 |
| 64 | SSB | CEA, FE | 168.79696 | 1247.6196 | 1966 |
| 65 | SSB | CEA, FE | 170.36528 | 1482.5471 | 1967 |
| 66 | SSB | CEA, FE | 192.62868 | 1746.4223 | 1968 |
| 67 | SSB | CEA, FE | 210.26197 | 2010.8458 | 1969 |
| 68 | SSB | CEA, FE | 208.87197 | 2122.3618 | 1970 |
| 69 | SSB | CEA, FE | 217.68088 | 2250.4911 | 1971 |

| | | | | |
|---|---|---|---|---|
| 70 | SSB CEA, FE | 201.71375 | 2139.8496 | 1972 |
| 71 | SSB CEA, FE | 192.23652 | 1916.2478 | 1973 |
| 72 | SSB CEA, FE | 177.27196 | 1538.3620 | 1974 |
| 73 | SSB CEA, FE | 167.26824 | 1408.1008 | 1975 |
| 74 | SSB CEA, FE | 126.52876 | 1216.5327 | 1976 |
| 75 | SSB CEA, FE | 133.80915 | 1338.4520 | 1977 |
| 76 | SSB CEA, FE | 112.85780 | 1143.4891 | 1978 |
| 77 | SSB CEA, FE | 93.93469 | 1064.0355 | 1979 |
| 78 | SSB CEA, FE | 85.76932 | 1282.6114 | 1980 |
| 79 | SSB CEA, FE | 80.63457 | 1947.5565 | 1981 |
| 80 | SSB CEA, FE | 94.77074 | 2703.1509 | 1982 |
| 81 | SSB CEA, FE | 127.72116 | 3662.0451 | 1983 |
| 82 | SSB CEA, FE | 139.32211 | 3812.6062 | 1984 |
| 83 | SSB CEA, FE | 155.85742 | 4363.4659 | 1985 |
| 84 | SSB CEA, FE | 152.69610 | 4284.8762 | 1986 |
| 85 | SSB CEA, FE | 147.37058 | 4417.1832 | 1987 |
| 86 | SSB CEA, FE | 133.56934 | 4288.8246 | 1988 |
| 87 | SSB CEA, FE | 120.34325 | 3822.2585 | 1989 |
| 88 | SSB CEA, FE | 111.19063 | 3339.9248 | 1990 |
| 89 | SSB CEA, FE | 101.34997 | 2766.0956 | 1991 |
| 90 | SSB CEA, FE | 88.01350 | 2555.1920 | 1992 |
| 91 | SSB CEA, FE | 94.71360 | 3114.1209 | 1993 |
| 92 | SSB CEA, FE | 100.80089 | 3387.5588 | 1994 |
| 93 | SSB CEA, FE | 100.86929 | 3380.1055 | 1995 |
| 94 | SSB CEA, FE | 96.66224 | 3146.3550 | 1996 |
| 95 | SSB CEA, FE | 94.37793 | 2968.3560 | 1997 |
| 96 | SSB CEA, FE | 81.74393 | 2535.9808 | 1998 |
| 97 | SSB CEA, FE | 82.84627 | 2762.2338 | 1999 |
| 98 | SSB CEA, FE | 80.04559 | 2788.5809 | 2000 |
| 99 | SSB CEA, FE | 84.01575 | 2954.8660 | 2001 |
| 100 | SSB CEA, FE | 81.34538 | 2844.1541 | 2002 |
| 101 | SSB CEA, FE | 80.65924 | 2932.4907 | 2003 |
| 102 | SSB CEA, FE | 86.58040 | 3245.6868 | 2004 |
| 103 | SSB CEA, FE | 78.78259 | 2860.9129 | 2005 |
| 104 | SSB CEA, FE | 77.17262 | 2646.0761 | 2006 |
| 105 | SSB CEA, FE | 72.05430 | 2231.1689 | 2007 |
| 106 | SSB CEA, FE | 63.54837 | 1779.2715 | 2008 |
| 107 | SSB CEA, FE | 67.05085 | 1940.4768 | 2009 |
| 108 | SSB CEA, FE | 67.24250 | 2005.1880 | 2010 |
| 109 | SSB CEA, FE | 76.32899 | 2444.2387 | 2011 |
| 110 | SSB CEA, FE | 89.42754 | 2960.5153 | 2012 |
| 111 | SSB CEA, FE | 103.90316 | 3384.9907 | 2013 |
| 112 | SSB CEA, FE | 108.29424 | 3364.8423 | 2014 |

| | | | | | |
|---|---|---|---|---|---|
| 113 | SSB | CEA, FE | 99.67968 | 3122.6491 | 2015 |
| 114 | SSB | CEA, FE | 101.38400 | 3319.2674 | 2016 |
| 115 | SSB | CEA, FE | 111.89005 | 3690.6429 | 2017 |
| 116 | SSB | CEA, FE | 110.72711 | 3426.9770 | 2018 |
| 117 | SSB | CEA, FE | 114.82409 | 3255.5603 | 2019 |
| 118 | SSB | CEA, FE | 96.66492 | 2434.2877 | 2020 |
| 119 | SSB | CEA, FE | 92.21926 | 2468.2623 | 2021 |
| 120 | SSB | CEA, FE | 122.26997 | 3248.7539 | 2022 |
| 121 | SSB | CEA, FE | 133.16426 | 3161.3806 | 2023 |
| 122 | SSB | CEA, FE | 163.67251 | 3208.3073 | 2024 |
| 123 | Recruits | CEA, FE | 3111.66347 | 6694.6821 | 1964 |
| 124 | Recruits | CEA, FE | 6188.79694 | 27055.8182 | 1965 |
| 125 | Recruits | CEA, FE | 5189.02841 | 16000.6137 | 1966 |
| 126 | Recruits | CEA, FE | 6656.18057 | 26928.4899 | 1967 |
| 127 | Recruits | CEA, FE | 6390.37696 | 23195.5687 | 1968 |
| 128 | Recruits | CEA, FE | 6837.07603 | 27268.9306 | 1969 |
| 129 | Recruits | CEA, FE | 6221.93235 | 23875.3560 | 1970 |
| 130 | Recruits | CEA, FE | 4623.87084 | 14017.0964 | 1971 |
| 131 | Recruits | CEA, FE | 3553.84684 | 10203.9050 | 1972 |
| 132 | Recruits | CEA, FE | 4131.74442 | 21850.2979 | 1973 |
| 133 | Recruits | CEA, FE | 2891.72832 | 14740.7878 | 1974 |
| 134 | Recruits | CEA, FE | 2345.39781 | 13819.8700 | 1975 |
| 135 | Recruits | CEA, FE | 2014.66085 | 12742.5522 | 1976 |
| 136 | Recruits | CEA, FE | 1933.47535 | 14153.7468 | 1977 |
| 137 | Recruits | CEA, FE | 2269.58880 | 25413.7786 | 1978 |
| 138 | Recruits | CEA, FE | 3240.49173 | 61631.5070 | 1979 |
| 139 | Recruits | CEA, FE | 2294.81720 | 30340.7577 | 1980 |
| 140 | Recruits | CEA, FE | 2501.79434 | 36111.8063 | 1981 |
| 141 | Recruits | CEA, FE | 1793.35744 | 18429.3440 | 1982 |
| 142 | Recruits | CEA, FE | 3015.25402 | 54972.2840 | 1983 |
| 143 | Recruits | CEA, FE | 1600.61054 | 16107.1135 | 1984 |
| 144 | Recruits | CEA, FE | 2296.13318 | 40008.8483 | 1985 |
| 145 | Recruits | CEA, FE | 1305.08445 | 15074.7616 | 1986 |
| 146 | Recruits | CEA, FE | 853.80918 | 8559.7067 | 1987 |
| 147 | Recruits | CEA, FE | 671.56176 | 6674.0336 | 1988 |
| 148 | Recruits | CEA, FE | 866.97932 | 12380.1168 | 1989 |
| 149 | Recruits | CEA, FE | 1626.03031 | 49835.3664 | 1990 |
| 150 | Recruits | CEA, FE | 1106.57871 | 24019.0193 | 1991 |
| 151 | Recruits | CEA, FE | 976.31035 | 19027.8505 | 1992 |
| 152 | Recruits | CEA, FE | 1477.46135 | 39486.7150 | 1993 |
| 153 | Recruits | CEA, FE | 834.55396 | 14520.0837 | 1994 |
| 154 | Recruits | CEA, FE | 710.98814 | 11061.9600 | 1995 |
| 155 | Recruits | CEA, FE | 1035.77701 | 23565.6881 | 1996 |

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156 Recruits CEA, FE   1224.52357 33309.8411 1997
157 Recruits CEA, FE    824.11162 16331.7217 1998
158 Recruits CEA, FE    841.03967 17633.9557 1999
159 Recruits CEA, FE   1005.41332 25969.7543 2000
160 Recruits CEA, FE   1133.77260 35364.9076 2001
161 Recruits CEA, FE    869.70466 23649.8979 2002
162 Recruits CEA, FE    661.98546 15523.5766 2003
163 Recruits CEA, FE    418.22239  7097.6097 2004
164 Recruits CEA, FE    359.40357  5185.5070 2005
165 Recruits CEA, FE    629.09387 13055.5610 2006
166 Recruits CEA, FE   1098.51578 29384.2252 2007
167 Recruits CEA, FE    750.16343 13836.9683 2008
168 Recruits CEA, FE   1823.59914 55238.2687 2009
169 Recruits CEA, FE   1158.77034 25804.3838 2010
170 Recruits CEA, FE    860.66650 15731.1723 2011
171 Recruits CEA, FE    721.43589 12880.5225 2012
172 Recruits CEA, FE   1477.07553 46372.1722 2013
173 Recruits CEA, FE   1597.04792 49874.1413 2014
174 Recruits CEA, FE    985.79624 20194.7800 2015
175 Recruits CEA, FE    562.01442  7100.5741 2016
176 Recruits CEA, FE    591.25944  6864.9109 2017
177 Recruits CEA, FE    925.68818 12837.3220 2018
178 Recruits CEA, FE   3215.10320 74603.0707 2019
179 Recruits CEA, FE   1853.93433 20275.1711 2020
180 Recruits CEA, FE   1849.65860 13255.9447 2021
181 Recruits CEA, FE   2066.01367 10978.7076 2022
182 Recruits CEA, FE   3576.00025 16263.4065 2023
183 Recruits CEA, FE  14510.48069 51371.3228 2024
```

## Other models in development

The "Woods Hole Assessment model" (WHAM) is written in TMB and is a fully state-space model (Stock and Miller (2021)). Initial trials are available on the github repository.

The Rceattle model is developed by the AFSC and is written in TMB (Adams et al. (2022)). The model is an extension of the ADMB version used for multi-species model in the Eastern Bering Sea (Jurado-Molina, Livingston, and Ianelli (2005), Holsman et al. (2024)) and has a number of newer features. It is generalizable to include a variety of data types using lengths and can be set up for species tracked for sexual dimorphic growth and other processes. Since it has been recast into TMB, it allows for the ability to specify random effects in a number of processes. One development of the model is to have the flexibility to bridge among extant

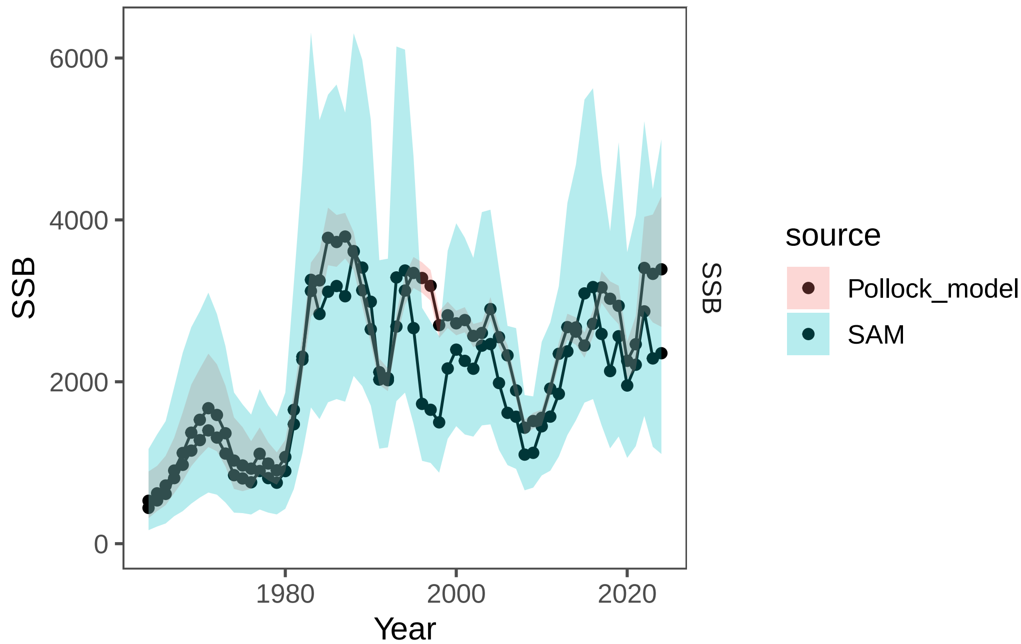


Figure 28: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

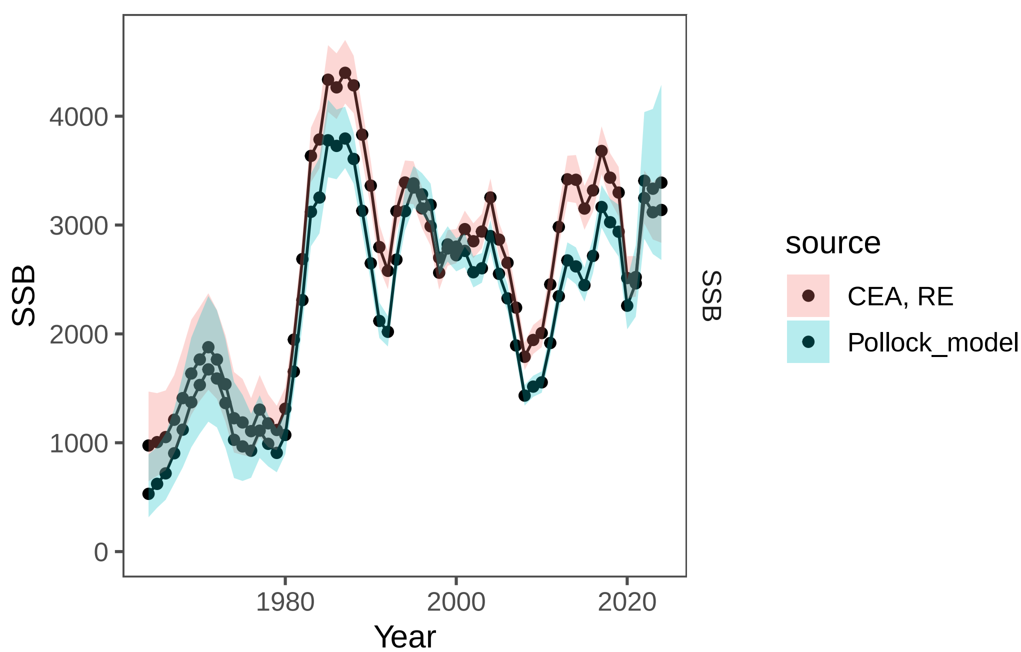


Figure 29: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

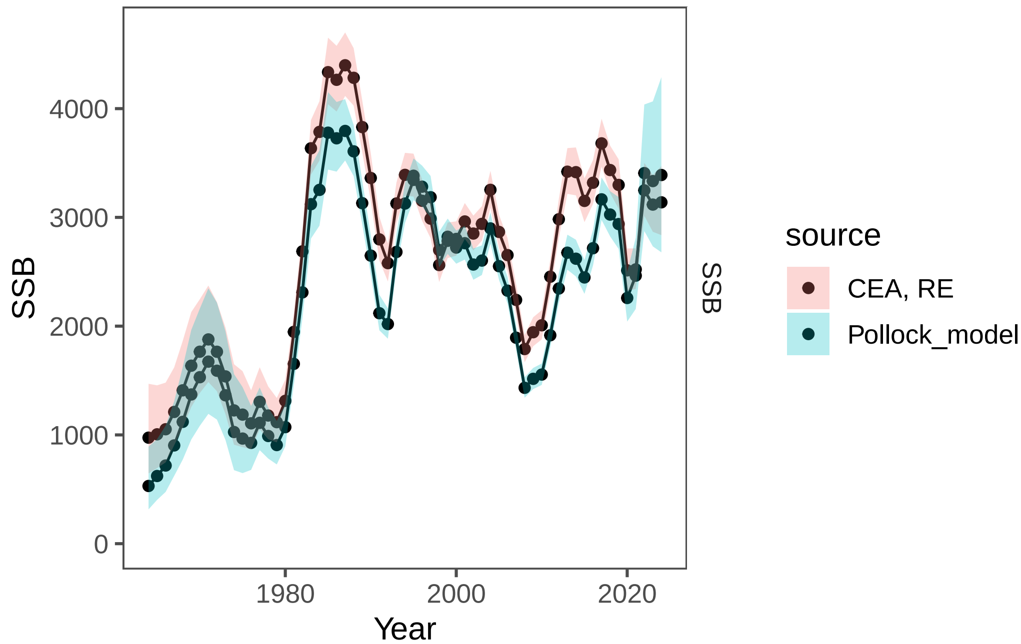


Figure 30: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

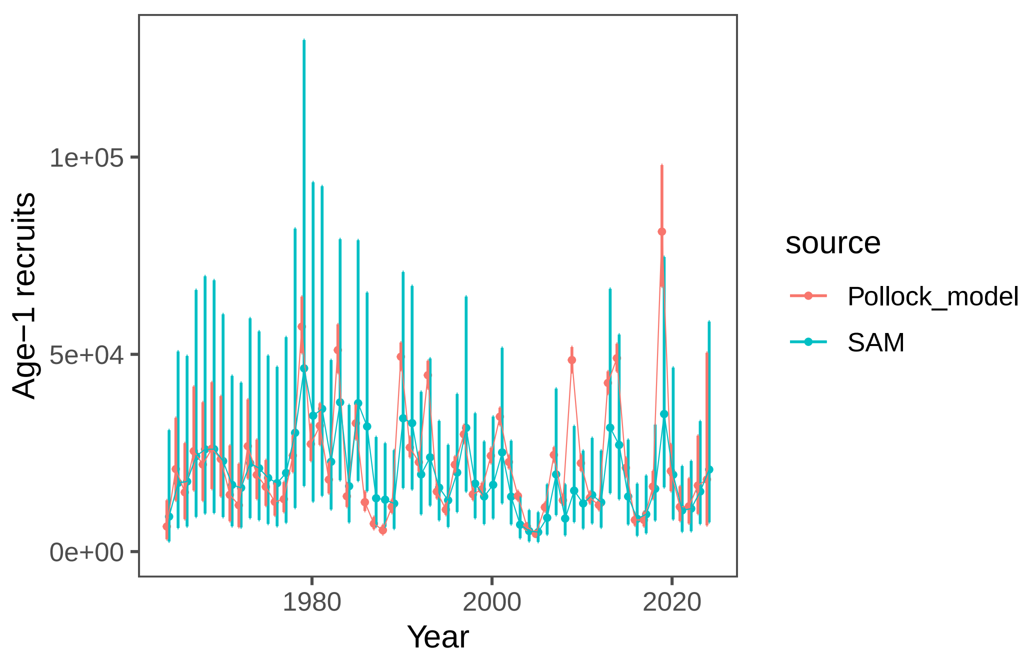


Figure 31: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

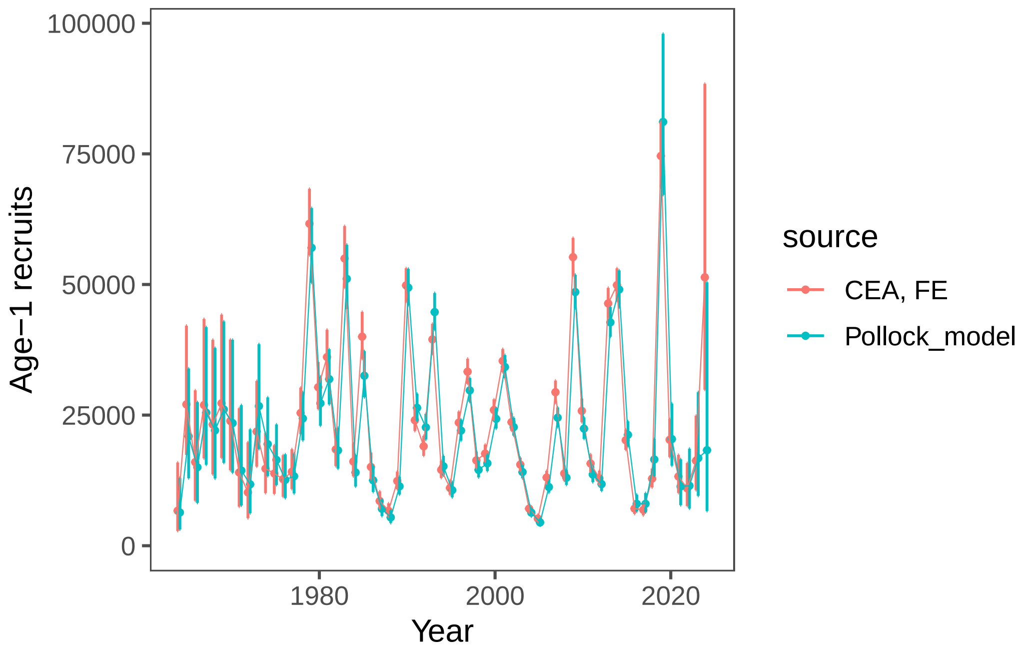


Figure 32: Model results comparing last year's model with the SAM model for spawning biomass (top) and recruitment at age 1 (bottom).

single-species modeling platforms by using it in "single-species" mode. To illustrate that, we introduce an application of Rceattle to the EBS pollock data set. The model is not yet fully developed, but we present preliminary results for consideration. Initial runs are favorable but further work is needed to bridge the selectivity options.

## Spatial modeling of the EBS pollock stock

In development is a spatially disaggregated general model by colleagues at the Auke Bay lab and the University of Alaska, Fairbanks (M. Cheng (2025)). This general model which has flexibility to specify regions and a variety of options including random effects. While this model is not yet fully developed, we present preliminary results here. The model is written in RTMB and is similar to the WHAM model. We pursue this to explore the potential for a more flexible model that can be used for more explicit considerations of stocks linked to different regions (e.g., the US and Russian portions of the EBS). Also, this model to add sex-specificity and random effects (M. L. H. Cheng et al. (2025)).

Figure 36 shows the model results comparing time-varying selectivity whereas Figure 37 shows the model results time series. The difference in the age-1 recruits (Figure 38) is likely due to the fact that the spatial model currently has constant M-at-age compared to the base model.

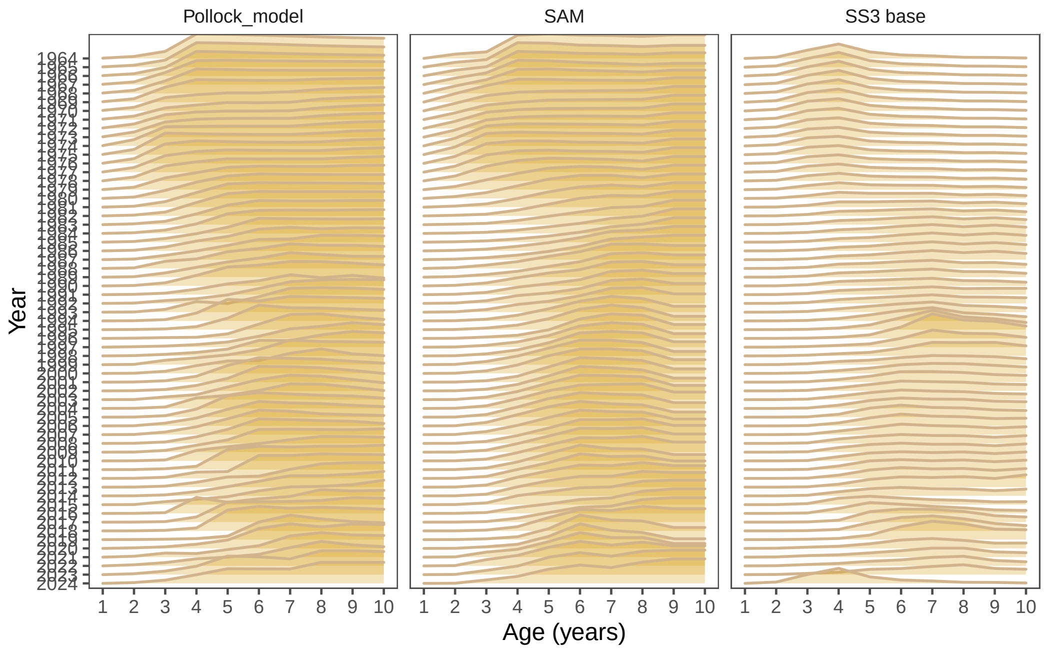


Figure 33: Model results comparing last year's model with the Rceattle model.

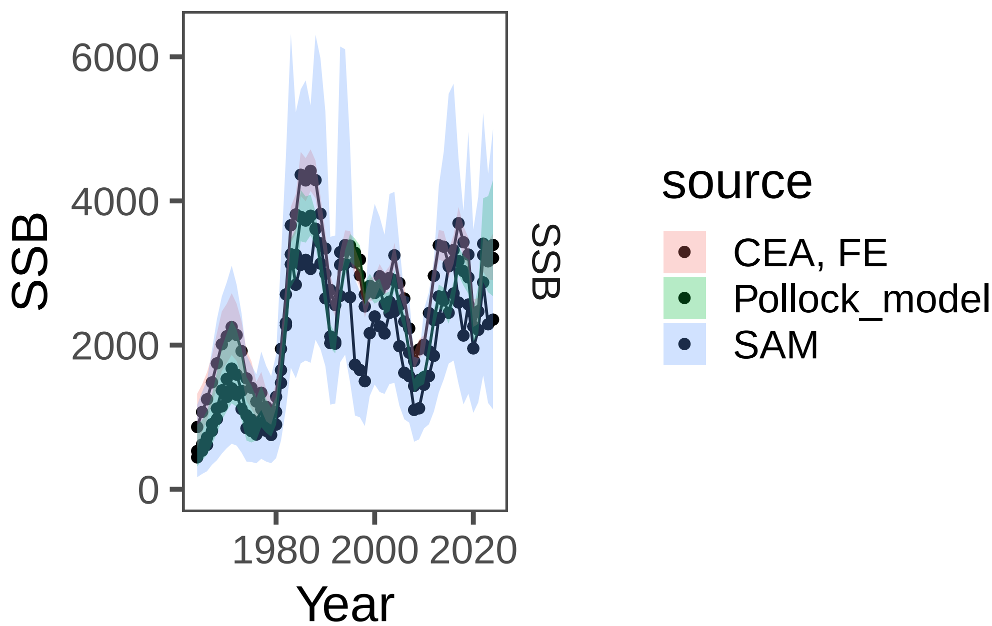


Figure 34: Model results comparing last year's model with the Rceattle model.

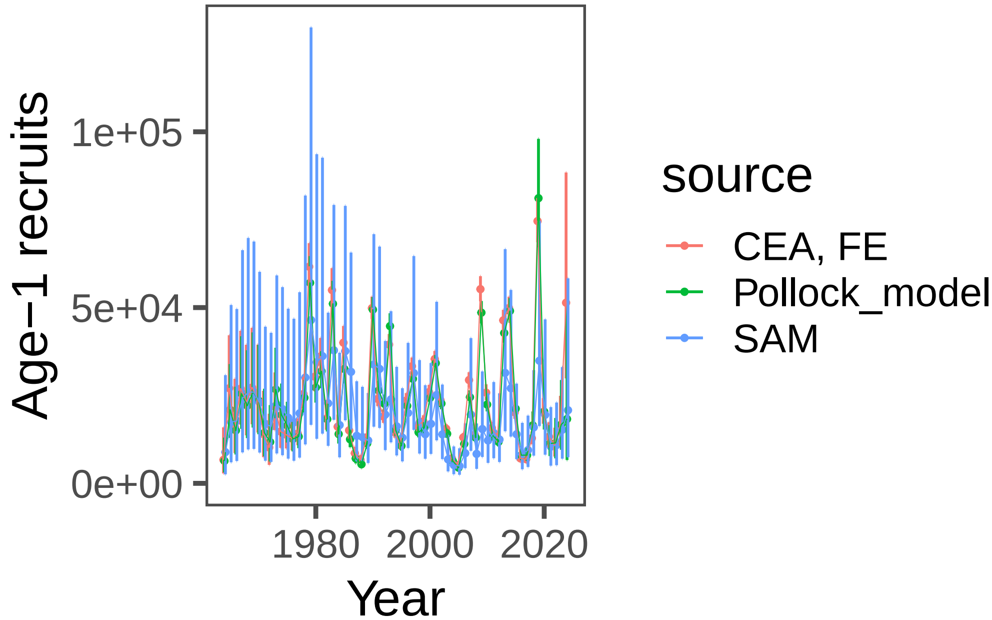


Figure 35: Model results comparing last year's model with the Rceattle model.

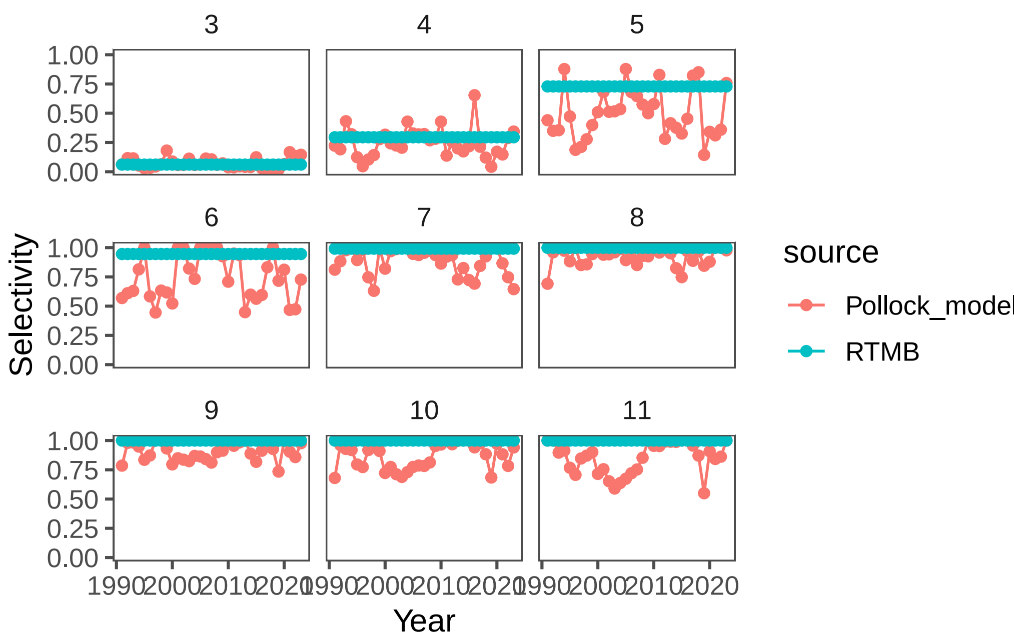


Figure 36: Model results comparing last year's model with the SPock model.

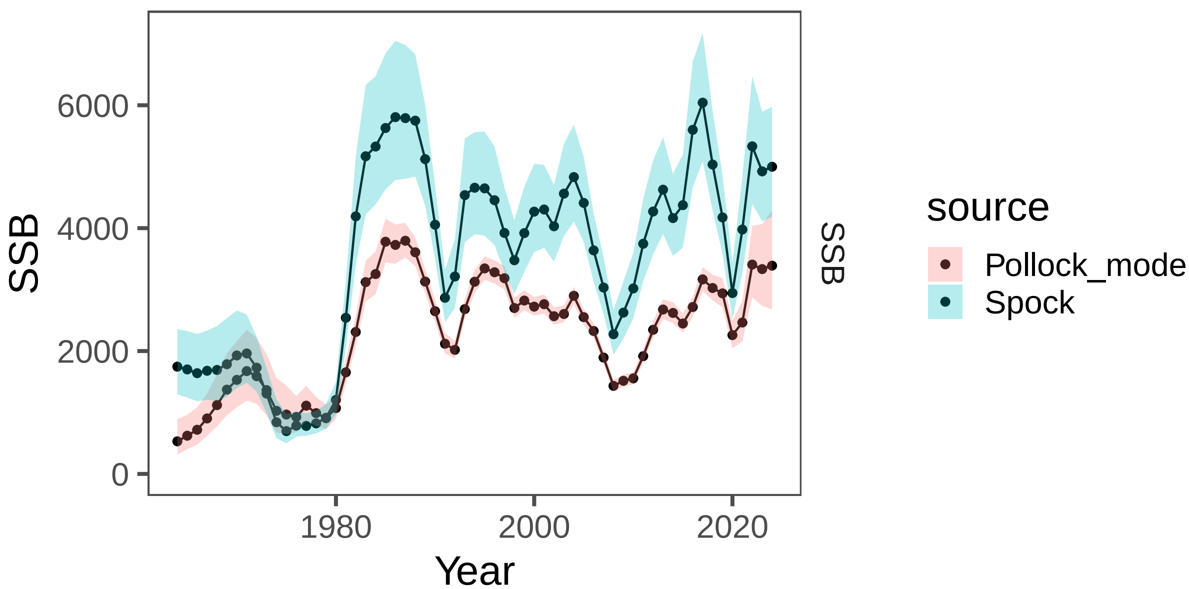


Figure 37: Model results comparing last year's model with the SPock model.

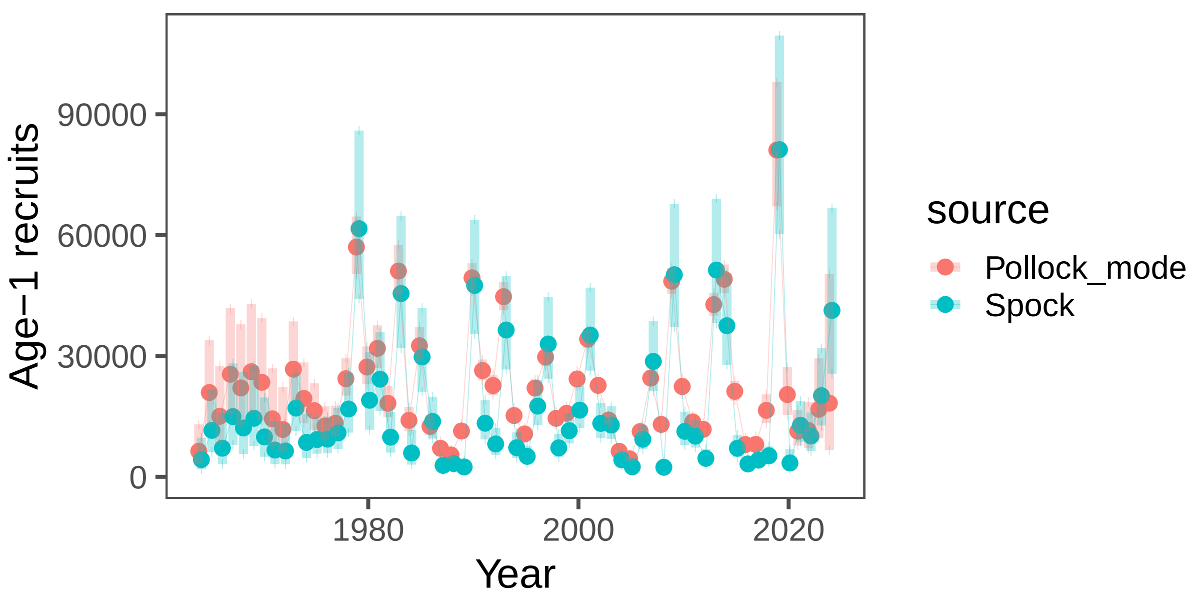


Figure 38: Model results comparing last year's model with the SPock model.

# Bottom-trawl survey spatial distribution modeling

## Abundance Index Methods

For model-based indices in the Bering Sea, we fitted observations of numerical abundance or biomass per unit area (where the choice of abundance or biomass varied by stock at the request of assessors) from all sampling locations in the 83-112 bottom trawl survey of the EBS, 1982-2025, including:

- Exploratory northern extension samples in 2001, 2005, and 2006
- 83-112 samples available in the NBS in 1982, 1985, 1988, 1991, 2010, 2017-2019, 2021-2023, and 2025

NBS samples collected prior to 2010 and in 2018 did not follow the 20 nautical mile sampling grid that was used in other survey years. Assimilating these unbalanced data requires extrapolating into unsampled areas, facilitated by including a spatially varying response to cold-pool extent (Thorson 2019). This spatially varying coefficient term was estimated for both linear predictors of a delta-model, and detailed comparison of results for EBS pollock has shown that it has a small but notable effect on these indices and resulting stock assessment outputs (O'Leary et al. 2020).

Rather than using the cold-pool covariate for yellowfin sole, we instead used the mean bottom temperature within the inner and middle domain strata from an interpolated temperature product. All models were fitted in the sdmTMB R package (https://pbs-assess.github.io/sdmTMB; Anderson et al. (2022)). All environmental data used as covariates were computed within the coldpool R package (https://github.com/afsc-gap-products/coldpool; Rohan, Barnett, and Charriere (2022)).

We used a spatiotemporal Poisson-link delta-model (Thorson 2018) involving two linear predictors and a gamma distribution to model positive catch rates. We predicted population density across the entire EBS and NBS in each year, using AFSC GAP-approved prediction grids. Spatial and spatiotemporal fields were estimated using a spatial mesh including 250 "knots" whose locations were determined using a K-means algorithm to place the knots evenly over space, in proportion to the dimensions of the prediction grid.

While the prior VAST model was fitted with a 750 knot mesh, we found that sdmTMB models fitted with mesh resolutions from 250-750 knots all provided extremely similar index estimates and that reducing the number of knots dramatically improved computation time. We estimated geometric anisotropy (how spatial autocorrelation declines with differing rates over distance in some cardinal directions than others) and included a spatial and spatiotemporal term for both linear predictors.

To facilitate prediction of density in regions with unsampled years, we specified that the spatiotemporal fields were structured over time as an AR(1) process (where the magnitude of autocorrelation was estimated as a fixed effect for each linear predictor). However, in cases where the AR(1) correlation parameter  was estimated to be close to 1 for either model component, the model would be collapsed into a simpler structure by specifying  = 1, i.e., modeling spatiotemporal variation as a random walk. We did not include any temporal correlation for intercepts, which we treated as fixed effects for each linear predictor and year. Finally, we used epsilon bias-correction to correct for retransformation bias (Thorson and Kristensen 2016).

We checked model fits for convergence by confirming that the derivative of the marginal likelihood with respect to each fixed effect was sufficiently small (less than ~ 0.001) and that the Hessian matrix was positive definite, along with other model checks produced by the function `sdmTMB::sanity()`. We then checked goodness of model fit by computing Dunn-Smyth randomized quantile residuals (Dunn and Smyth 1996) and visualizing these using a quantile-quantile plot within the DHARMa R package (Hartig 2021). We also evaluated the distribution of these residuals over space in each year, and inspected them for evidence of residual spatiotemporal patterns. Results compared favorably among methods, and the model index result was very similar to previous estimates (Figure 39).

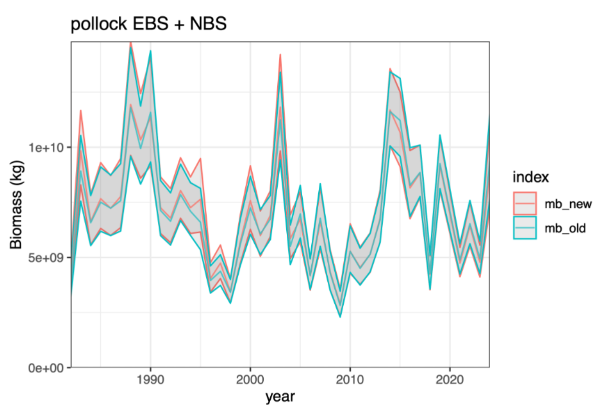


Figure 39: Model results comparing last year's model with the revised sdmTMB model.

## Model-based Age Composition Methods

For model-based estimation of age compositions in the Bering Sea, we fitted observations of numerical abundance-at-age at each sampling location. This was made possible by applying a year-specific, region-specific (EBS and NBS) age-length key to records of numerical abundance and length composition. In subcategories (combinations of year, length, age, sex) that contained insufficient data, age composition was computed from length composition given a globally pooled age-length key.

We computed these estimates in the tinyVAST R package (https://vast-lib.github.io/tinyVAST; Thorson et al. (2024)), assuming a Poisson-link delta-model (Thorson 2018) involving two linear predictors, and a gamma distribution to model positive catch rates. We did not include any density covariates in estimation of age composition for consistency with models used in the previous assessments and due to computational limitations.

We estimated the spatial range assuming geometric isotropy (i.e, spatial autocorrelation declines with equal rates over distance in all cardinal directions). We used the same extrapolation grid as implemented for abundance indices, but here we modeled spatial and spatiotemporal fields with a mesh with a coarser spatial resolution than the index model, here using 50 "knots". This reduction in the spatial resolution of the model, relative to that used for the abundance indices, was necessary due to the increased computational load of fitting multiple age categories.

The estimates of abundance at age were bias-corrected with the same method as the abundance index. We implemented the same diagnostics to check convergence and model fit as those used for the abundance index. Results compared favorably among methods (Figure 40).

Both the model-based bottom trawl index and age composition estimates were calculated using code in the AFSC GAP model-based indices GitHub repository (https://github.com/afsc-gap-products/model-based-indices). Design-based estimates of bottom trawl products were calculated using code in the gapindex R package (https://github.com/afsc-gap-products/gapindex).

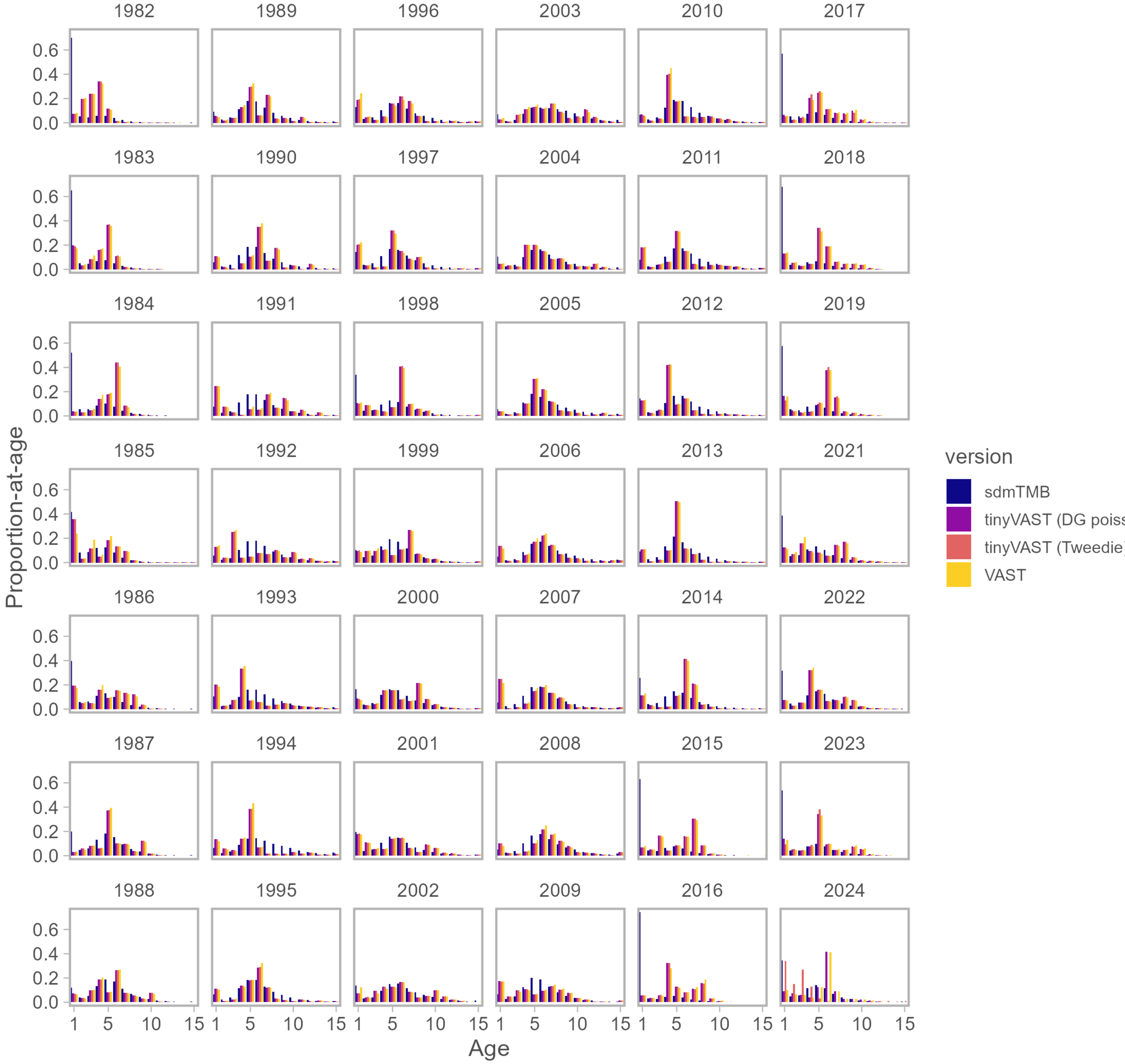


Figure 40: Model results comparing last year's model with the revised sdmTMB model.

# Pollock model fishery selectivity notes

The following provides more detail based on the actual code as adopted from Butterworth, Ianelli, and Hilborn (2003).

## Initial Setup

### Average Selectivity

$$\overline{sel} = \log\left(\text{mean}(\exp(\mathbf{c}))\right)$$

where $\mathbf{c}$ is the vector of base selectivity coefficients.

### Base Year Selectivity (t = stsel)

For ages 1 to nsel:

$$\log(S_{t,a}) = c_a \quad \text{for } a = 1, 2, ..., nsel$$

For older ages (plateau assumption):

$$\log(S_{t,a}) = c_{nsel} \quad \text{for } a = nsel + 1, ..., nages$$

### Initial Normalization

$$\log(S_{t,a}) = \log(S_{t,a}) - \log\left(\frac{1}{nages}\sum_{a=1}^{nages}\exp(\log(S_{t,a}))\right)$$

This ensures: $\prod_{a=1}^{nages} S_{t,a}^{1/nages} = 1$

## Time-Varying Component

### Selectivity Evolution (t > stsel)

For years without regime changes:

$$\log(S_{t+1,a}) = \log(S_{t,a}) \quad \text{for all ages}$$

For years with regime changes (when $t+1 \in$ yrs_ch_fsh):

**Ages 1 to nsel:**

$$\log(S_{t+1,a}) = \log(S_{t,a}) + \delta_{ii,a} \quad \text{for } a = 1, 2, ..., nsel$$

**Older ages (plateau maintained):**

$$\log(S_{t+1,a}) = \log(S_{t+1,nsel}) \quad \text{for } a = nsel + 1, ..., nages$$

where: - $\delta_{ii,a}$ is the selectivity deviation for change event $ii$ and age $a$ - $ii$ is the index of the current selectivity change event

### Annual Normalization

For each year $t$:

$$\log(S_{t,a}) = \log(S_{t,a}) - \log\left(\frac{1}{nages}\sum_{a=1}^{nages}\exp(\log(S_{t,a}))\right)$$

and the actual selectivity is:

$$S_{t,a} = \exp(\log(S_{t,a}))$$

## Constraints and Properties

1. **Normalization Constraint**:

$$\prod_{a=1}^{nages} S_{t,a}^{1/nages} = 1 \quad \forall t$$

2. **Dome-shaped Constraint**:

$$S_{t,a} = S_{t,nsel} \quad \text{for } a > nsel$$

3. **Continuity**: Selectivity changes only occur in specified years (`yrs_ch_fsh`)
4. **Deviation Structure**:

$$\delta_{ii} \sim \text{estimated deviations with penalties}$$

## Implementation Notes

- Base coefficients $\mathbf{c}$ are estimated parameters
- Deviations $\delta$ are typically penalized to prevent overfitting
- The plateau assumption prevents unrealistic selectivity patterns for older ages
- Geometric mean normalization maintains interpretability relative to fishing mortality

# Regularization and Penalty Terms

The selectivity model includes several penalty terms to ensure realistic and stable estimates: Penalty Term 1: Deviation Magnitude Penalty $P_1 = \frac{\lambda_1}{N_{groups}} \sum_{i=1}^{N_{changes}} \sum_{a=1}^{nsel} \delta_{i,a}^2$ where:

$\lambda_1$ = ctrl_flag(10) (penalty weight parameter) $N_{groups}$ = group_num_fsh (number of fishery groups) This shrinks deviations toward zero when not informed by data

Penalty Term 2: Base Selectivity Smoothness Penalty $P_2 = \frac{\lambda_2}{N_{changes}} \sum_{a=1}^{nsel-2} \left[\nabla^2 \log(S_{styr,a})\right]^2$ where:

$\lambda_2$ = ctrl_flag(11) (smoothness penalty weight) $\nabla^2 \log(S_{t,a}) = \log(S_{t,a+2}) - 2\log(S_{t,a+1}) + \log(S_{t,a})$ (second difference) This penalizes high curvature in the base selectivity pattern

Penalty Term 3: Change Year Smoothness Penalty $P_3 = \frac{\lambda_2}{N_{changes}} \sum_{i=1}^{N_{changes}} \sum_{a=1}^{nsel-2} \left[\nabla^2 \log(S_{t_i,a})\right]^2$ where $t_i$ represents each change year. This maintains smooth selectivity curves in all change years. Penalty Term 4: Temporal Stability Penalty (Random Walk) $P_4 = \sum_{i=1}^{N_{changes}} \frac{1}{2\sigma_i^2} \sum_{a=1}^{nages} \left[\log(S_{t_i-1,a}) - \log(S_{t_i,a})\right]^2$ where:

$\sigma_i$ = sel_ch_sig_fsh(i) (change magnitude parameter for change event $i$) This penalizes large year-to-year changes in selectivity Only applied when $t_i \neq styr$ (change year is not the start year)

Total Penalty Function $P_{total} = P_1 + P_2 + P_3 + P_4$ The total negative log-likelihood becomes: $-\log L = -\log L_{data} + P_{total}$ Penalty Interpretation

$P_1$: Prevents overfitting by shrinking deviations toward zero $P_2$ and $P_3$: Enforce smooth, realistic selectivity curves (penalize jagged patterns) $P_4$: Constrains temporal evolution (selectivity changes smoothly over time)

The penalty weights ($\lambda_1$, $\lambda_2$) and variance parameters ($\sigma_i$) control the balance between fitting data and maintaining realistic selectivity patterns. Implementation Notes

Base coefficients **c** are estimated parameters Deviations $\delta$ are penalized to prevent overfitting The plateau assumption prevents unrealistic selectivity patterns for older ages Geometric mean normalization maintains interpretability relative to fishing mortality Second differences approximate curvature (second derivatives) for smoothness penalties

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